Master's Thesis

Riccati-based Primal Regularizations in Interior-Point Solvers for Indefinite Quadratic Programs

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Chapter 1

Introduction

1.1 Motivation

In the field of nonlinear model predictive control (NMPC), in each sampling period, a nonlinear optimal control problem (OCP) is solved to compute the optimal control input for the current state of the system. In order to solve the continuous-time optimal control problems, direct multiple shooting [1] is a widely used method, which transcribes the OCP into a nonlinear programming problem (NLP) by discretizing the time horizon into multiple shooting intervals [2].

Sequential Quadratic Programming (SQP) provides a suitable approach for solving the NLP in real-time practice, which linearizes at each iterate to generate a QP subproblem. In the real-time iteration [3], one SQP iteration is performed per sampling time, warm-started at the shifted previous solution and linearized before the new state is known, so that only the QP solve remains between measurement and feedback. The solving of the QP subproblem therefore determines the feedback delay, and it can be solved by structure-exploiting solver such as HPIPM [4], which runs Interior Point Method (IPM) for solving QPs

Nevertheless, when the full Hessian of the QP subproblems is provided by the exact Hessian of the Lagrangian [5], it could be indefinite, thus indicates a nonconvex problem. The structure-exploiting solvers embedded in the SQP, such as qpDUNES [6], FORCES [7] and HPIPM, however, fails to solve these QPs, for they require convex QPs with positive definite (PD) full Hessian. Though FORCES and HPIPM can in principle handle the nonconvex QPs with indefinite full Hessian as long as the reduced Hessian, i.e. the Hessian on the null space of the dynamics, is PD, but the code optimization of both solvers still requires a PD full Hessian, thus convex QPs are intended by the solvers instead of nonconvex

QPs. One solution to use these solvers in this situation is to regularize at the SQP level, i.e. to regularize the full Hessian of the QP before it is passed to the solver, which in general modifies the solution of the subproblems and in the real-time iteration the feedback itself.

Given the Hessian on the null space of the dynamics and the active constraints is PD, CONVEXITY [8] can avoid the modification when the full Hessian is not PD. It applies a backward sweep similar to Riccati recursion to shift curvature between neighboring stages, and adds curvature with a tunable penalty on the estimated active constraints outside the null space of the dynamics, which regularizes the full Hessian without touching the reduced Hessian. And the resulting QP keeps the same primal solution if the estimation is correct. In the case where both Hessian matrices are indefinite, it falls back to SQP level regularization and thereby modifies the QP by replacing the negative eigenvalues of the indefinite full Hessian stage-wisely by small positive values.

In an interior point method, however, the barrier term of the inequality constraints adds this curvature by itself. Since the curvature contribution increases on the constraints that is becoming active and vanishes on the inactive ones, estimation on the active set and tuning on the penalty weight can be avoided. Especially when the iterates are close enough to the solution on the constraint, the barrier term can make the reduced Hessian PD without the solver knowing which constraints are active. In the early iterations, or where this condition fails, the Hessian still has to be corrected. If a correction inside the solver is made after the barrier curvature has been added, unlike a correction at the SQP level, it only has to supply the curvature that the barrier lacks and changes only the Newton step, while the residuals, and hence the found solution, remain those of the original QP.

A structure-exploiting interior point QP solver that handles indefiniteness itself is therefore interesting as an alternative to regularization at the SQP level. This leaves open how the correction should be made inside the Riccati recursion. This thesis addresses this question for indefinite QPs with optimal control structure.

1.2 Contributions

The main contributions of this thesis are the following:

1. **A regularization inside the Riccati recursion** of an interior point QP

solver, which corrects the Hessian only at the stages where it is needed and is inactive near the solution, together with an improved stage-wise strategy (Chapter 3).

2. **A benchmark of** 175 **indefinite OCP QPs** collected from the SQP iterations of eight OCPs, with an analysis of where the indefiniteness comes from (Chapter 4).
3. **A numerical comparison** on this benchmark, where the methods are compared for performance from the efficiency and estimated computation perspectives, and an analysis of solving failures is presented. (Section 5.4).

1.3 Related open-source solvers

This thesis builds on two open-source software packages for embedded optimal control, acados and HPIPM.

acados [9] is a modular framework for fast NMPC. It transcribes the OCP by direct multiple shooting, solves the resulting NLP by SQP or its real-time iteration variant, and passes the QP subproblems to one of several interfaced QP solvers. When the exact Hessian is used, acados handles indefiniteness at the SQP level: the QP Hessian can be regularized before the QP is passed to the solver or be replaced by a Gauss–Newton approximation. In this thesis, acados is used to generate the SQP iterates from which the benchmark QPs are collected (Section 4.1).

HPIPM [4] is a primal–dual interior point solver for OCP-structured QPs and the default QP solver in acados. It solves the KKT system of each iteration by Riccati recursion, with a cost linear in the horizon length, and relies on high-performance linear algebra routines tailored to the small and medium-sized matrices of optimal control. HPIPM assumes a convex QP: the Riccati recursion requires the reduced Hessian blocks to be positive definite, so it fails on indefinite QPs. This failure is precisely the criterion by which the benchmark QPs are collected (Section 4.1), and the prototype solver used in the experiments reproduces the iteration behavior of HPIPM on convex QPs, so that the regularization is the only difference between the two (Section 5.2).

1.4 Outline

The remainder of this thesis is organized as follows. Chapter 2 introduces the primal–dual interior point method for OCP-structured QPs, the Riccati recursion used to solve its KKT system, and the barrier update and step size control strategies adopted in the solver. Chapter 3 classifies primal regularization by its target matrix, develops the Riccati-based primal regularization, derives the improved stage regularization strategy from the failure of minimum perturbation, and proves that the regularization is asymptotically inactive. Chapter 4 constructs the benchmark of indefinite QPs from the SQP iterations of eight OCPs and analyzes how the indefiniteness is distributed over the stages and where it originates. Chapter 5 describes the prototype solver and the experiment setup, compares the Riccati-based methods with full Hessian regularization and inertia correction on the benchmark, and discusses the instances on which the methods fail. Finally, Chapter 6 summarizes the results and outlines directions for future work.

Chapter 2

Primal–dual interior point method for optimal control problems

In this chapter, the theoretical background regarding relevant techniques regarding IPM are introduced. The primal-dual interior point method (IPM) is introduced first in Section 2.1 as the core algorithm for solving the optimal control problems in this thesis. Then, the Riccati recursion is introduced for efficiently solving the KKT system of the optimal control problem in Section 2.1. Finally, the adaptive barrier update and step size control strategies are discussed in Section 2.3 and Section 2.4 considering the nonconvex feature in this work.

2.1 Primal–dual interior point method

Consider a QP with equality and inequality constraints, the formulation can be written as:

$$\begin{aligned} \underset{y}{\text{minimize}} \quad & \frac{1}{2} y^\top H y + g^\top y \\ \text{subject to} \quad & Fy = b, \\ & Gy \leq d, \end{aligned} \tag{2.1}$$

Where y are the decision variables, H represents the full Hessian, g is the linear term, the F and G stand for the linear constraint matrices for equality and inequality. Since the constraint are linear w.r.t y, the definiteness of the full Hessian matrix H distinguishes the QP problem (2.1) as convex or nonconvex

By introducing slack variables t, we have the following formulation:

$$\begin{aligned} \underset{y,\,t}{\text{minimize}} \quad & \frac{1}{2} y^\top H y + g^\top y \\ \text{subject to} \quad & Fy = b, \\ & Gy + t = d, \\ & t \geq 0, \end{aligned} \tag{2.2}$$

we can further obtain the Lagrangian function of the QP problem (2.2) as

$$\mathcal{L}(y, t, \pi, \lambda) = \frac{1}{2} y^\top H y + g^\top y + \pi^\top (Fy - b) + \lambda^\top (Gy + t - d), \tag{2.3}$$

where π and $\lambda \geq 0$ are the Lagrange multipliers associated with the equality and inequality constraints, respectively. Then the KKT conditions of the QP problem (2.2) are given by

$$Hy + g + F^\top \pi + G^\top \lambda = 0, \tag{2.4a}$$

$$Fy - b = 0, \tag{2.4b}$$

$$Gy + t - d = 0, \tag{2.4c}$$

$$\lambda_j \, t_j = 0, \quad j = 1, \ldots, n_c, \tag{2.4d}$$

$$\lambda \geq 0, \quad t \geq 0. \tag{2.4e}$$

where n_c is the number of inequality constraints. In primal-dual IPM, a homotopy parameter τ, also called as barrier parameter, is applied to the only nonlinear equation (2.4d) for relaxing the complementarity, the equation (2.4d) can be rewritten as:

$$\lambda_j \, t_j = \tau, \quad j = 1, \ldots, n_c, \tag{2.5}$$

the parameter τ is shrunk toward zero as the iterations approach the solution with certain strategy, and size step control is applied to ensure the satisfaction of inequality (2.4e). The corresponding update strategy and step size control will be discussed in the following sections.

In order to solve the nonlinear system of equations (2.4), the Newton's method is applied to the linearized system at each iteration i to obtain the step

$(\Delta y, \Delta\pi, \Delta\lambda, \Delta t)$, which takes the form of:

$$\begin{bmatrix} H & F^\top & G^\top & 0 \\ F & 0 & 0 & 0 \\ G & 0 & 0 & I \\ 0 & 0 & T_i & \Lambda_i \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta\pi \\ \Delta\lambda \\ \Delta t \end{bmatrix} = - \begin{bmatrix} r^i_{\text{stat}} \\ r^i_{\text{eq}} \\ r^i_{\text{ineq}} \\ r^i_{\text{comp}} \end{bmatrix}, \tag{2.6}$$

where the diagonal matrices $T_i = \text{diag}(t_i)$, $\Lambda_i = \text{diag}(\lambda_i)$, and the perturbed KKT residuals follow as:

$$\begin{bmatrix} r^i_{\text{stat}} \\ r^i_{\text{eq}} \\ r^i_{\text{ineq}} \\ r^i_{\text{comp}} \end{bmatrix} := \begin{bmatrix} Hy_i + g + F^\top\pi_i + G^\top\lambda_i \\ Fy_i - b \\ Gy_i + t_i - d \\ T_i\Lambda_i\mathbf{1} - \tau_i\mathbf{1} \end{bmatrix}, \tag{2.7}$$

Based on the structure of the KKT system (2.6), Schur complement reduction can be applied to reduce the system (2.6) to a saddle point problem, the symmetric system can be presented as:

$$\begin{bmatrix} \tilde{H}_i & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta\pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}^i_{\text{stat}} \\ r^i_{\text{eq}} \end{bmatrix}, \tag{2.8}$$

with $\tilde{H}_i = H + G^\top T_i^{-1}\Lambda_i G \succeq H$ and $\tilde{r}^i_{\text{stat}} = r^i_{\text{stat}} + G^\top T_i^{-1}(\Lambda_i r^i_{\text{ineq}} - r^i_{\text{comp}})$. Notably, the barrier term $G^\top T_i^{-1}\Lambda_i G$ is regarded as the curve contribution from the constraints, of which the weight on each constraint, i.e. the corresponding diagonal entry of $T_i^{-1}\Lambda_i$, increases on the constraints becoming active and tends to zero on the inactive ones, and the term is positive semidefinite since the iterates are kept strictly positive in t and λ with (2.4e).

In the (2.8), assuming F has full row rank, the null space method can be applied to reduce the system. Let the columns of Z form a basis of the null space of F, i.e. $FZ = 0$, and let the columns of Y span the range of $F^\top$, so that $[\,Y \;\; Z\,]$ is nonsingular and FY is invertible.

The primal step can then be decomposed uniquely as $\Delta y = Y\,\Delta y_y + Z\,\Delta y_z$, where only the range-space component enters the equality constraints, since $F\Delta y = FY\,\Delta y_y$. This yields the system where the reduced Hessian is exposed:

$$\tilde{M}_i\,\Delta y_z = -\Big(Z^\top\tilde{r}^i_{\text{stat}} + Z^\top\tilde{H}_i\,Y\,\Delta y_y\Big), \tag{2.9}$$

where the equality Jacobian F is eliminated by $Z^\top$ (with $Z^\top F^\top = 0$), and feasibility fixes the range-space component $\Delta y_y = -(FY)^{-1} r^i_{\text{eq}}$. And we can have the augmented reduced Hessian $\tilde{M}_i = Z^\top \tilde{H}_i Z$ and the right-hand side $Z^\top \tilde{r}^i_{\text{stat}} + Z^\top \tilde{H}_i Y \Delta y_y$ shown in this formulation.

2.2 Efficient solution with Riccati recursion

Within the context of optimal control problems, the QP (2.1) can be detailed as:

$$\begin{aligned}
\underset{x,\,u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \left(\frac{1}{2} \begin{bmatrix} x_k \\ u_k \end{bmatrix}^\top \begin{bmatrix} Q_k & S_k^\top \\ S_k & R_k \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \begin{bmatrix} g_x^k \\ g_u^k \end{bmatrix}^\top \begin{bmatrix} x_k \\ u_k \end{bmatrix} \right) \\
& + \frac{1}{2} x_N^\top Q_N x_N + (g_x^N)^\top x_N \\
\text{subject to} \quad & x_0 = \hat{x}_0, \\
& x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, \ldots, N-1, \\
& G_x^k x_k + G_u^k u_k \le d_k, \qquad\quad k = 0, \ldots, N-1, \\
& G_x^N x_N \le d_N,
\end{aligned} \tag{2.10}$$

where $x_k \in \mathbb{R}^{n_x}$ and $u_k \in \mathbb{R}^{n_u}$ are the state and control at stage k, and N is the horizon length. $Q_k \in \mathbb{R}^{n_x \times n_x}$, $R_k \in \mathbb{R}^{n_u \times n_u}$, and $S_k \in \mathbb{R}^{n_u \times n_x}$ are the state, control, and state–control coupling blocks of the stage Hessian H_k, $g_x^k \in \mathbb{R}^{n_x}$ and $g_u^k \in \mathbb{R}^{n_u}$ are the corresponding gradients, and Q_N, g_x^N are the terminal Hessian and gradient. $A_k \in \mathbb{R}^{n_x \times n_x}$ and $B_k \in \mathbb{R}^{n_x \times n_u}$ are the Jacobians of the discretized dynamics with respect to the state and control, $b_k \in \mathbb{R}^{n_x}$ is the dynamics residual, and $\hat{x}_0 \in \mathbb{R}^{n_x}$ is the initial state. $G_x^k \in \mathbb{R}^{n_c^k \times n_x}$ and $G_u^k \in \mathbb{R}^{n_c^k \times n_u}$ are the Jacobians of the n_c^k inequality constraints at stage k, including linear constraints and box constraints on states and controls, with right-hand side $d_k \in \mathbb{R}^{n_c^k}$.

Following same induction in Section 2.1, we obtain the KKT system of (2.10), which exhibits a tridiagonal structure due to the temporal nature of the problem, as:

$$\begin{bmatrix}
\ddots & & & & & \\
& 0 & -I & & & \\
& -I & \tilde{Q}_k & \tilde{S}_k^\top & A_k^\top & \\
& 0 & \tilde{S}_k & \tilde{R}_k & B_k^\top & \\
& 0 & A_k & B_k & 0 & -I \\
& & & & -I & \ddots
\end{bmatrix}
\begin{bmatrix} \vdots \\ \Delta\pi_k \\ \Delta x_k \\ \Delta u_k \\ \Delta\pi_{k+1} \\ \vdots \end{bmatrix}
= -
\begin{bmatrix} \vdots \\ r_{\text{eq}}^{k-1} \\ \tilde{r}_{\text{stat},x}^k \\ \tilde{r}_{\text{stat},u}^k \\ r_{\text{eq}}^k \\ \vdots \end{bmatrix}, \tag{2.11}$$

the augmented block matrices $\tilde{Q}_k$, $\tilde{R}_k$, and $\tilde{S}_k$ contains the barrier term as:

$$\begin{aligned}\tilde{Q}_k &= Q_k + (G_x^k)^\top \Lambda_k T_k^{-1} G_x^k,\\ \tilde{R}_k &= R_k + (G_u^k)^\top \Lambda_k T_k^{-1} G_u^k,\\ \tilde{S}_k &= S_k + (G_x^k)^\top \Lambda_k T_k^{-1} G_u^k,\end{aligned}$$

This tridiagonal structure allows for efficient solution of the KKT system using Riccati recursion [10], which is consisted of a backward and forward pass. The backward pass computes the feedback gains K and cost-to-go matrices P, while the forward pass computes the optimal state and control trajectories, the algorithmic details of the Riccati recursion can be summarized as follows:

Algorithm 1 Riccati Recursion for Solving OCP Structure QP

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\text{stat},x}$

// Backward recursion
for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$
  $K_k \leftarrow -\tilde{M}_k^{-1} \Gamma_k$
  $k_k \leftarrow -\tilde{M}_k^{-1} \left(\tilde{r}^k_{\text{stat},u} + B_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1})\right)$
  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k$
  $p_k \leftarrow \tilde{r}^k_{\text{stat},x} + A_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

// Forward recursion
for $k = 0$ **to** $N-1$ **do**
  $\Delta u_k \leftarrow K_k \Delta x_k + k_k$
  $\Delta x_{k+1} \leftarrow A_k \Delta x_k + B_k \Delta u_k + r^k_{\text{eq}}$
  $\Delta \pi_{k+1} \leftarrow P_{k+1} \Delta x_{k+1} + p_{k+1}$
  $\Delta t_k \leftarrow -(G_x^k \Delta x_k + G_u^k \Delta u_k + r^k_{\text{ineq}})$
  $\Delta \lambda_k \leftarrow T_k^{-1}(-r^k_{\text{comp}} - \Lambda_k \Delta t_k)$
end for

By applying the Riccati recursion, we solve the KKT system (2.9) without computing the inverse of the reduced Hessian $\tilde{M}_i$, and the computational complexity is reduced from $\mathcal{O}(N^3)$ to $\mathcal{O}(N)$, thus achieving an efficient solving by exploiting the OCP structure.

2.3 Adaptive barrier update

In this section, two closely related adaptive barrier update strategies are discussed for the primal-dual IPM. While the monotone decrease of the barrier parameter τ, such as the Fiacco-McCormick scheme [11], enjoys strong convergence guarantees, it may lead to slow convergence and practical limitations [12]. We therefore prefer a more aggressive, adaptive update of τ to achieve faster convergence, which is the focus of this section

2.3.1 Mehrotra's predictor-corrector

For IPM in convex QPs, the Mehrotra's predictor–corrector [13] scheme is usually considered as the default choice for fast convergence, which adaptively updates the barrier parameter τ and adds a second-order corrector for improving the direction, at essentially the cost of one extra back-solve. In the strategy, the barrier parameter τ is defined as:

$$\tau_i = \sigma_i \mu_i, \qquad \mu_i = \frac{t_i^\top \lambda_i}{n_c}, \tag{2.12}$$

where μ_i is the duality measure, and $\sigma_i \geq 0$, typically in $[0, 1]$, is the centering parameter that controls the trade-off between centering and aggressive steps. And in each iteration, two solves of (2.6) that share the same left-hand side and differ only in the complementarity right-hand side are performed, which involves two steps: predictor step (affine scaling step), and corrector step.

Predictor Step. The strategy starts with computing a predictor step, the linear system (2.6) is solved with $\sigma_i = 0$ in the complementarity residual r^i_{comp} and no correction considered, i.e. the resulting complementarity residual is written as:

$$r^i_{\text{comp,aff}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1} = T_i \Lambda_i \mathbf{1}, \tag{2.13}$$

to obtain the affine scaling direction $(\Delta y_{\text{aff}}, \Delta \pi_{\text{aff}}, \Delta \lambda_{\text{aff}}, \Delta t_{\text{aff}})$. The affine step is used to estimate the centering parameter σ_i for the next step.

Centering. After the predictor step is obtained, The centering parameter σ is updated based on the estimated duality measure μ_{aff} after the affine step, which

is defined as:

$$\sigma_i = (\mu_{\text{aff}}/\mu_i)^3 \tag{2.14}$$

$$\mu_{\text{aff}} = \frac{(t_i + \alpha^i_{\text{prim,aff}}\Delta t_{\text{aff}})^\top(\lambda_i + \alpha^i_{\text{dual,aff}}\Delta\lambda_{\text{aff}})}{n_c}, \tag{2.15}$$

where the step size $\alpha^i_{\text{prim,aff}}$ and $\alpha^i_{\text{dual,aff}}$ are determined by ensuring that the updated slack variables and Lagrange multipliers remain nonnegative, i.e.

$$\alpha^i_{\text{prim,aff}} = \min\left(1, \min_{j:\Delta t_{\text{aff},j}<0} -\frac{t^i_j}{\Delta t_{\text{aff},j}}\right), \tag{2.16}$$

$$\alpha^i_{\text{dual,aff}} = \min\left(1, \min_{j:\Delta\lambda_{\text{aff},j}<0} -\frac{\lambda^i_j}{\Delta\lambda_{\text{aff},j}}\right). \tag{2.17}$$

If $\mu_{\text{aff}} \ll \mu_i$, the affine step achieves a substantial reduction in the duality gap, which sharply gives $\sigma_i \to 0$ (near-Newton, aggressive), while a stalled step ($\mu_{\text{aff}} \approx \mu_i$) gives $\sigma_i \to 1$ (recentering toward the central path).

Corrector Step. With the updated centering parameter σ_i and calculated affine steps, the corrector step is computed by solving the linear system (2.6) again. In the corrector step, the complementarity residual is modified to include a second-order correction term that accounts for the nonlinear effects of the affine step, which is given by:

$$r^i_{\text{comp,cor}} = T_i\Lambda_i\mathbf{1} + \Delta T_{\text{aff}}\Delta\Lambda_{\text{aff}}\mathbf{1} - \sigma_i\mu_i\mathbf{1}, \tag{2.18}$$

where, $\Delta T_{\text{aff}} = \text{diag}(\Delta t_{\text{aff}})$, $\Delta\Lambda_{\text{aff}} = \text{diag}(\Delta\lambda_{\text{aff}})$, the term $\Delta T_{\text{aff}}\Delta\Lambda_{\text{aff}}\mathbf{1}$ cancels the second-order error that the affine step leaves in $t_j\lambda_j$, and $-\sigma_i\mu_i\mathbf{1}$ supplies the centering.

This second solve returns the final direction $(\Delta y, \Delta\pi, \Delta\lambda, \Delta t)$ directly, a step size control will be followed to ensure the positivity of the slack variables and Lagrange multipliers.

Condition on Corrector Step. In practice, the corrector step is usually applied conditionally, the mechanism is, therefore, called conditional Mehrotra's predictor-corrector step [12], where the corrected direction is only accepted if it

does not largely increase the duality measure compared with the affine step, i.e.,

$$\mu_{\text{cor}} = \frac{(t_i + \alpha^i_{\text{prim,cor}} \Delta t)^\top (\lambda_i + \alpha^i_{\text{dual,cor}} \Delta \lambda)}{n_c} \leq 2\mu_{\text{aff}}, \tag{2.19}$$

where $\alpha^i_{\text{prim,cor}}$ and $\alpha^i_{\text{dual,cor}}$ are computed as the affine step sizes, but along the corrected direction. Otherwise, (2.6) is solved again without the correction term.

2.3.2 Mehrotra's probing

However, Mehrotra's predictor-corrector scheme is not necessarily the best choice in the nonconvex case. The corrector adds $\Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1}$ to predict the complementarity error of a full affine step, which assumes the affine direction is an accurate Newton step. In nonconvex QPs, the direction has to be modified to stay a descent direction, so it is inexact and often much longer than the Newton step.

And since the correction is quadratic in the affine direction, this error is then squared, and the step size control in the method often cuts the step size to almost nothing. The conditional Mehrotra's predictor-corrector step can still miss this, because when the steps are short, $\mu_{\text{cor}} \approx \mu_{\text{aff}} \approx \mu_i$, so $\mu_{\text{cor}} \leq 2\mu_{\text{aff}}$ holds and the corrected direction is accepted.

As mentioned in [12], the Mehrotra's probing strategy can be applied alternatively to the predictor-corrector scheme, which taking a more conservative approach to update the barrier parameter τ by discarding the corrector step in predictor-corrector scheme but keeps the centering step. The corresponding complementarity residual is given by:

$$r^i_{\text{comp,pro}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1}, \tag{2.20}$$

where the centering parameter σ_i is updated using (2.14) after the affine step, as in the predictor–corrector scheme. Probing uses the affine step only through $\sigma = (\mu_{\text{aff}}/\mu)^3$, so an inexact affine step just gives $\sigma \approx 1$, i.e. a centering step.

2.4 Step size control

In this section, we discuss the step size control employed in the primal-dual interior point solver. The fraction-to-the-boundary (FTB) rule is first introduced as a common strategy for step size control in IPM, which ensures that the updated

slack variables and Lagrange multipliers remain positive. Then, a two-phase line search with backtracking is applied to ensure progress in either the objective function or the constraint violation, particularly in nonconvex cases where the FTB rule may not be sufficient to guarantee convergence.

2.4.1 Fraction-to-the-boundary rule

The fraction-to-the-boundary (FTB) rule is a common strategy for step size control in primal-dual interior point methods. The maximal step size $\alpha^i_{\text{prim,max}}$ and $\alpha^i_{\text{dual,max}}$ are computed without violating the nonnegativity of the slack variables and Lagrange multipliers, respectively, as:

$$\alpha^i_{\text{prim,max}} = \min_{j:\Delta t_j<0} -\frac{t^i_j}{\Delta t_j}, \tag{2.21}$$

$$\alpha^i_{\text{dual,max}} = \min_{j:\Delta \lambda_j<0} -\frac{\lambda^i_j}{\Delta \lambda_j}. \tag{2.22}$$

Then, in practice, the actual step sizes are chosen as a fraction of the maximum step sizes, i.e.,

$$\alpha^i_{\text{prim}} = \min(1, \eta\alpha^i_{\text{prim,max}}), \tag{2.23}$$

$$\alpha^i_{\text{dual}} = \min(1, \eta\alpha^i_{\text{dual,max}}), \tag{2.24}$$

where $\eta \in (0, 1)$ is a parameter that controls the fraction of the maximum step size to be taken, typically set to a value close to 1 to ensure sufficient progress while maintaining feasibility.

2.4.2 Two-phase line search with backtracking

In nonconvex cases, the FTB rule may not be sufficient to guarantee convergence, as the search direction with FTB-controlled step size may not descend.

A two-phase line search inspired by the funnel method [14] is applied on top of the FTB step size. Denoting by $\theta_i = \|c_i\|$ the constraint violation at the current iterate and by $\rho_i = \nabla f_i^\top d$ the directional derivative of the objective along the search direction d, the *switching condition*:

$$\alpha^i_{\text{prim}}(-\rho_i) > \kappa_{\text{s}}\,\theta_i^2, \qquad \kappa_{\text{s}} > 0, \tag{2.25}$$

decides which of the two phases the trial step belongs to. Because (2.25) involves only quantities of the current iterate and the trial step size, it is evaluated *before* a candidate is formed.

If (2.25) fails, the predicted objective decrease is small relative to the constraint violation, so objective descent is not yet a meaningful measure of progress and the iterate is in the *h-phase*. The full FTB step is then accepted without any merit evaluation. This is justified by the structure of the subproblem: the constraints of the QP are affine and the Newton direction solves the corresponding rows of the linear system exactly, so $\theta(\alpha^i_{\text{prim}}) = (1 - \alpha^i_{\text{prim}})\,\theta_i$ holds exactly along the step. The constraint violation therefore decreases monotonically in α^i_{prim}, and the largest admissible step gives the fastest feasibility progress.

If (2.25) holds, sufficient objective progress is expected and the iterate is in the *f-phase*. A candidate iterate $z^+(\alpha^i_{\text{prim}})$ is formed in the primal variables (x, u, s), its objective f^+ is evaluated, and the step is accepted when the Armijo's condition [15]

$$f^+ \leq f_i + \gamma\,\alpha^i_{\text{prim}}\,\rho_i, \qquad \gamma \in (0, 1), \tag{2.26}$$

is satisfied.

Otherwise the step size is contracted, $\alpha^i_{\text{prim}} \leftarrow \beta\,\alpha^i_{\text{prim}}$ with $\beta \in (0, 1)$, and the search repeats. Contraction shrinks the left-hand side of (2.25), so a trial that starts in the f-phase may switch to the h-phase after a few backtracks and terminate there. If all $n_{\max}$ trials are rejected, the search reverts to the original FTB step size. The dual step size α^i_{dual} is left untouched throughout, as the merit function reads only the primal variables. The procedure is summarized in Algorithm 2.

Two simplifications distinguish this scheme from the funnel method of [14].

First, no funnel bound $\theta_{\max}$ is maintained. In the funnel method, a trial iterate is admissible only if it stays inside the relaxed feasible set $\{z : \theta(z) \leq \theta_{\max}\}$, and the bound is contracted whenever a feasibility step is accepted. Here the constraints are affine, so $\theta(\alpha^i_{\text{prim}}) = (1 - \alpha^i_{\text{prim}})\,\theta_i \leq \theta_i$ for every $\alpha^i_{\text{prim}} \in (0, 1]$: no trial step can increase the constraint violation, the funnel membership test is satisfied by construction and carries no information, and the h-phase acceptance test and the funnel update become redundant.

Second, the switching condition (2.25) is built on the first-order term $-\nabla f_i^\top d$ alone, in line with the Armijo condition [15], whereas [14] uses the second-order model reduction $\Delta m_f = -\frac{1}{2} d^\top H d - \nabla f_i^\top d$. The reason is the nonconvexity of the

subproblem: since H is generally indefinite, the sign of $d^\top H d$ is not controlled, so the second-order contribution to Δm_f may be misleading rather than informative for classifying the step. To compensate for dropping the second-order term, the constant κ_s is set to a small value, so that objective-reducing steps are not misclassified as h-phase steps.

Algorithm 2 Two-phase line search

Input: iterate z_i, direction d, FTB steps $\alpha^i_{\mathrm{prim}}, \alpha^i_{\mathrm{dual}}$, f_i, $\theta_i = \|c_i\|$, $\rho_i = \nabla f_i^\top d$
$\alpha^i_{\mathrm{prim,FTB}} \leftarrow \alpha^i_{\mathrm{prim}}$
for $\mathrm{ls} = 1$ **to** $n_{\max}$ **do**
  if $\alpha^i_{\mathrm{prim}}(-\rho_i) \le \kappa_\mathrm{s}\,\theta_i^2$ **then** ▷ h-phase: feasibility dominates
    break ▷ accept FTB-scaled step
  end if
  form candidate $z^+(\alpha^i_{\mathrm{prim}})$
  compute f^+
  if $f^+ \le f_i + \gamma\,\alpha^i_{\mathrm{prim}}\,\rho_i$ **then** ▷ f-phase: Armijo
    break
  end if
  if $\mathrm{ls} = n_{\max}$ **then**
    $\alpha^i_{\mathrm{prim}} \leftarrow \alpha^i_{\mathrm{prim,FTB}}$ ▷ all trials rejected: revert to the FTB step
  else
    $\alpha^i_{\mathrm{prim}} \leftarrow \beta\,\alpha^i_{\mathrm{prim}}$
  end if
end for
return $\alpha^i_{\mathrm{prim}},\ \alpha^i_{\mathrm{dual}}$

Chapter 3

Primal regularization for indefinite quadratic programs

In this chapter, we turn our attention to QPs with an indefinite Hessian matrix, namely indefinite QPs, and we focus on the primal regularization for the indefinite QPs. A primal regularization can be regarded by three choices: how the correction is computed, when it is applied, and where it is applied. Section 3.1 starts with a review of the numerical strategies to compute the correction, and then propose the reduced Hessian as the right target for the detection and the regularization. Section 3.2 shows that the Riccati recursion provides the reduced Hessian implicitly, and develops the Riccati-based primal regularization, which detects the indefiniteness and applies the correction stagewise inside the recursion. Section 3.3 analyses why the minimum-perturbation rule fails in this setting, and derives an improved stage regularization strategy from that analysis. Section 3.4 concludes the chapter by proving that, owing to the detection on the reduced Hessian, the regularization becomes asymptotically inactive as the iterates approach the solution.

3.1 Primal regularization

In indefinite QPs, the regularization needs to be applied to the Hessian matrix to ensure a descent search direction in the interior point method. The resulting regularized reduced KKT system can be generally presented as:

$$\begin{bmatrix} \tilde{H}_i + E_i & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta \pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}^i_{\mathrm{stat}} \\ r^i_{\mathrm{eq}} \end{bmatrix}, \tag{3.1}$$

where E_i is the adjustment matrix that renders the Hessian positive definite at the i-th iteration. This process is referred to as primal regularization.

Three choices are left open by this formulation: how the adjustment E_i is computed, when it is applied, i.e. on which matrix the indefiniteness is detected, and where it is applied, i.e. which matrix it is derived from and applied to. The first is common ground, since the same numerical strategies apply whichever matrix is targeted. Section 3.1.1 therefore introduces these strategies first, after which Section 3.1.2 and Section 3.1.3 discuss when and where to regularize, each on the full or the reduced Hessian.

3.1.1 Regularization strategies

Three families of strategies are commonly used to construct the adjustment matrix E_i, which are Eigenvalue modification (EM), Modified Cholesky factorization (MC), Levenberg–Marquardt (LM) [16–18]. From these families we select four methods to be implemented in the solver: PROJECT and MIRROR from EM [8], GMW from MC [19], and inertia correction from LM [20].

Assume a matrix A admits the eigenvalue decomposition $A = VD_EV^\top$ and the Cholesky factorization $A = LD_IL^\top$, where V is an orthogonal matrix of eigenvectors, L is a unit lower triangular matrix, D_E is a diagonal matrix of real eigenvalues, and D_I is a diagonal matrix whose signs give the inertia of A. In this notation, the four methods read as follows:

PROJECT. This approach clips any negative or near-zero eigenvalues to a small positive constant $\epsilon > 0$, which keeps a minimal adjustment to the original matrix for positive definiteness, thus the minimal Frobenius norm of the adjustment matrix E [16], but may lead to an increase in the condition number if the original eigenvalues were not close to zero. The operation can be described as:

$$\text{project}(A, \epsilon) := V \max(\epsilon I, D_E)V^\top.$$

Where the max operator is applied component-wise.

MIRROR. This method reflects negative eigenvalues to their positive counterparts. It preserves the magnitude of the curvature in all directions, which often provides better scaling, but may result in a large modification to the original

matrix. The operation can be described as:

$$\text{mirror}(A, \epsilon) := V \max(\epsilon I, |D_E|) V^\top.$$

GMW. Instead of a spectral decomposition, the factorization is carried out in place, and each diagonal pivot is raised, whenever necessary, to keep it above ϵ and to bound the magnitude of the corresponding column of L [19]. Bounding the pivots from below therefore also bounds the condition number of the factor, and the accumulated increments form a nonnegative diagonal correction E with $A + E = LD_I L^\top$. This avoids the cost of an eigendecomposition and still guarantees positive definiteness, but the resulting E is generally not minimal and tends to over-modify A. The operation can be described as:

$$\text{gmw}(A, \epsilon) := LD_I L^\top, \qquad D_I \succeq \epsilon I.$$

Inertia Correction. A uniform multiple of the identity is added to shift every eigenvalue by the same amount. The shift avoids the cost of a spectral decomposition and is easy to implement, however, because it acts equally in all directions, it can degrade the scaling when only a few directions are indefinite. The operation can be described as:

$$\text{ic}(A, \Delta) := A + \Delta I, \qquad \Delta \geq 0.$$

In practice $\lambda_{\min}$ is not computed directly, and Δ is found by repeatedly attempting the $\text{LDL}^\top$ factorization and increasing Δ until every diagonal entry of D_I is positive.

3.1.2 Indefiniteness detection on reduced Hessian

A positive definite Hessian H makes the QP strictly convex and is therefore sufficient for a unique solution. It is also required by common structure-exploiting QP solvers in optimal control, such as qpDUNES [6], FORCES [7] and HPIPM [4].

An equality-constrained QP, however, only requires the reduced Hessian to be positive definite, according to Theorem 1, which is mentioned in [8] (for proof, see [16]):

Theorem 1. *Assume that LICQ holds. A QP with only equality constraints has*

a unique global optimum if and only if the reduced Hessian is positive definite:

$$M = Z^\top H Z \succ 0. \tag{3.2}$$

The positive definiteness of the reduced Hessian is a necessary and sufficient condition for the QP admitting a unique global optimum, whereas that of the full Hessian is only sufficient.

The Schur-complemented system (2.8) is the KKT system of the equality-constrained QP

$$\begin{aligned} \underset{\Delta y}{\text{minimize}} \quad & \frac{1}{2}\Delta y^\top \tilde{H}_i \Delta y + (\tilde{r}^i_{\text{stat}})^\top \Delta y \\ \text{subject to} \quad & F\Delta y + r^i_{\text{eq}} = 0. \end{aligned} \tag{3.3}$$

By Theorem 1, the step Δy is therefore the unique minimizer of this QP if and only if $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$. A detection of the indefiniteness of $\tilde{M}_i$ is therefore more preferable than that of $\tilde{H}_i$.

3.1.3 Regularization on full or reduced Hessian

Once the indefiniteness is detected, a correction is applied to the Hessian so that Δy is a descent direction. The correction can be derived from and applied to either the full or the reduced Hessian.

The full Hessian regularization takes the full Hessian $\tilde{H}$ as the target. In optimal control, it is commonly applied stage-wise to the block $\tilde{H}_k$ of each stage k,

$$\tilde{H}^{\text{reg}}_k = \begin{bmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{bmatrix} + E_k \succ 0, \quad k = 0, 1, \ldots, N-1, \tag{3.4}$$

where E_k is a diagonal or a full matrix, depending on the strategy in Section 3.1.1. This is simple to implement but conservative: at an iterate with $r^i_{\text{eq}} = 0$, Δy lies in the null space of F, so only the curvature of $\tilde{H}$ within this subspace affects the step. Making $\tilde{H}$ positive definite also removes negative curvature outside it, where the step never goes, so the adjustment can be larger than necessary and moves Δy away from the Newton direction.

With a part from [16, Theorem 16.6], the following result shows that, when equation constraint is satisfied, i.e. at an iterate with $r^i_{\text{eq}} = 0$, positive curvature in the null space is both necessary and sufficient for a descent direction.

Theorem 2. *Assume that LICQ holds, $r^i_{\text{eq}} = 0$, and $\tilde{M}_i = Z^\top \tilde{H}_i Z$ is nonsingular.*

Then $(\tilde{r}^i_{\text{stat}})^\top \Delta y < 0$ *holds for every* $\tilde{r}^i_{\text{stat}}$ *with* $Z^\top \tilde{r}^i_{\text{stat}} \neq 0$ *if and only if* $\tilde{M}_i \succ 0$.

Proof. Since $r^i_{\text{eq}} = 0$, the range-space component $\Delta y_y = -(FY)^{-1} r^i_{\text{eq}}$ vanishes, so $\Delta y = Z \Delta y_z$ and (2.9) reduces to $\tilde{M}_i \Delta y_z = -Z^\top \tilde{r}^i_{\text{stat}}$. Hence

$$(\tilde{r}^i_{\text{stat}})^\top \Delta y = -(Z^\top \tilde{r}^i_{\text{stat}})^\top \tilde{M}_i^{-1} (Z^\top \tilde{r}^i_{\text{stat}}). \tag{3.5}$$

For sufficiency, if $\tilde{M}_i \succ 0$, then $\tilde{M}_i^{-1} \succ 0$, and the right-hand side of (3.5) is negative for every $Z^\top \tilde{r}^i_{\text{stat}} \neq 0$.

For necessity, we proceed by contraposition. Suppose that $\tilde{M}_i$ is not positive definite. Being nonsingular, it has an eigenpair (d_e, v) with $d_e < 0$. Since Z has full column rank, $\tilde{r}^i_{\text{stat}} = Z(Z^\top Z)^{-1} v$ satisfies $Z^\top \tilde{r}^i_{\text{stat}} = v \neq 0$, and (3.5) with $\tilde{M}_i^{-1} v = v/d_e$ gives $(\tilde{r}^i_{\text{stat}})^\top \Delta y = -v^\top v / d_e > 0$, so Δy is an ascent direction for this residual. □

A positive definite reduced Hessian is therefore exactly what a descent direction requires, and the reduced Hessian regularization, which takes $\tilde{M}_i$ as the target, enforces this condition.

However, $\tilde{M}_i$ is not directly available. Computing it requires a null-space basis of the equality constraints, e.g. via condensing in optimal control [21], which introduces extra computational cost for both the detection of Section 3.1.2 and the reduced Hessian regularization.

3.2 Riccati-based primal regularization

We thus propose a Riccati-based primal regularization method, which naturally obtains the reduced Hessian $\tilde{M}$ implicitly via Riccati recursion. The Riccati recursion serves two purposes. First, it detects the indefiniteness of the reduced Hessian $\tilde{M}$ with Cholesky factorization as the sign. Second, it allows the reduced Hessian $\tilde{M}$ to be regularized stage-wisely to ensure a positive definite reduced Hessian $\tilde{M}$, which is the target proposed in this work.

3.2.1 Block reduced Hessian from Riccati recursion

In this subsection, we will restate and establish the connection between the Riccati recursion and the block $\text{LDL}^\top$ factorization of reduced Hessian $\tilde{M}_i$ in (2.9), which not only provide a way to compute the reduced Hessian $\tilde{M}_i$ implicitly, but also a way to detect the indefiniteness of $\tilde{M}_i$.

From the perspective of the Riccati recursion, the backward pass can be concluded as follows:

Algorithm 3 Backward Pass of Riccati Recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\text{stat},x}$

for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}^k_{\text{stat},u} + B_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1})\right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}^k_{\text{stat},x} + A_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

We can find that $\tilde{M}_k = \tilde{R}_k + B_k^\top P_{k+1} B_k$ is exactly the block reduced Hessian obtained by Riccati recursion. And in practice the inverse of $\tilde{M}_k$ is not computed directly, instead, the Cholesky factorization of $\tilde{M}_k$ is computed for efficiency, which also serves as a detection of the indefiniteness of $\tilde{M}_k$.

From the perspective of the reduced Hessian $\tilde{M}_i$ in (2.9), eliminating the states through the dynamics leaves a dense reduced Hessian $\tilde{M}_i$ in the inputs $u_0, \ldots, u_{N-1}$, whose density stems from the coupling of any two stages via the state transition $\Phi_{i,j}$, defined as:

$$\Phi_{i,j} = \begin{cases} \prod_{k=j}^{i-1} A_k = A_{i-1} A_{i-2} \cdots A_j, & \text{if } i > j, \\ I, & \text{if } i = j. \end{cases}$$

The backward pass in Algorithm 3 is exactly a block $\text{LDL}^\top$ factorization of this $\tilde{M}_i$, carried out without ever forming it explicitly. The connection is made through the Schur complement: each cost-to-go matrix P_k is the Schur complement obtained by eliminating the states of the later stages, so that the pivot block $\tilde{M}_k$ is precisely the Schur complement of stage k in $\tilde{M}_i$ after all later stages have been eliminated. Consequently, the recursion $P_k = \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k$ is nothing but the successive formation of these Schur complements.

Collecting the pivots $\tilde{M}_{N-1}, \ldots, \tilde{M}_0$ into the block-diagonal factor D and eliminating the stages in the same order as the backward pass, from stage $N-1$

down to stage 0, the block $\mathrm{LDL}^\top$ factorization of the reduced Hessian $\tilde{M}_i$ can be expressed as:

$$\tilde{M}_i = L \begin{bmatrix} \tilde{M}_{N-1} & 0 & \cdots & 0 \\ 0 & \tilde{M}_{N-2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \tilde{M}_0 \end{bmatrix} L^\top$$

where the accumulated back-substitution coefficients, built from the feedback gains K_k and the transitions $\Phi_{i,j}$, form the unit lower-triangular factor L:

$$L = \begin{bmatrix} I & 0 & \cdots & 0 & 0 \\ -B_{N-2}^\top K_{N-1}^\top & I & \cdots & 0 & 0 \\ -B_{N-3}^\top \Phi_{N-1,N-2}^\top K_{N-1}^\top & -B_{N-3}^\top K_{N-2}^\top & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -B_0^\top \Phi_{N-1,1}^\top K_{N-1}^\top & -B_0^\top \Phi_{N-2,1}^\top K_{N-2}^\top & \cdots & -B_0^\top K_1^\top & I \end{bmatrix}.$$

This establishes the connection between positive definiteness of the reduced Hessian $\tilde{M}_i$ and the positive definiteness of the block reduced Hessians $\tilde{M}_k$ obtained from Riccati recursion. Since each pivot $\tilde{M}_k$ is a Schur complement of $\tilde{M}_i$ and L is nonsingular, the Cholesky factorizations of all $\tilde{M}_k$ in the backward pass succeed if and only if $\tilde{M}_i \succ 0$. We can therefore have the Riccati-based detection, as also used in Fatrop [22], where the Riccati recursion performs the detection on the reduced Hessian of Section 3.1.2 without forming $\tilde{M}_i$ and at no extra cost.

3.2.2 Stage regularization in Riccati recursion

With this connection, we can regularize the reduced Hessian $\tilde{M}_i$ without explicitly computing it, but by stage-wisely applying the regularization to the block reduced Hessian $\tilde{M}_k^i$ during the Riccati recursion. When the Cholesky factorization of $\tilde{M}_k^i$ fails, which indicates that the reduced Hessian $\tilde{M}_i$ is indefinite, we regularize the reduced Hessian block $\tilde{M}_k^i$ by adding a block adjust matrix E_k^i, i.e. $\tilde{M}_k^i + E_k^i$. The E_k^i can be computed from the strategies mentioned in 3.1.1. This gives a modification of the backward pass of Riccati recursion and can be concluded as follows:

Algorithm 4 Backward Pass of Modified Riccati Recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\text{stat},x}$, $\lambda > 0$

for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  if factorization fails **then**
    $\tilde{M}_k \leftarrow \tilde{M}_k + E_k$ ▷ Apply regularization
    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  end if
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}^k_{\text{stat},u} + B_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) \right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}^k_{\text{stat},x} + A_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

Since the block reduced Hessians $\tilde{M}_k$ are exactly the pivots of the block $\text{LDL}^\top$ factorization of $\tilde{M}_i$, enforcing the positive definiteness of every $\tilde{M}_k$ during the backward pass is equivalent to regularizing the reduced Hessian $\tilde{M}_i$ itself, and thus recovers the descent direction Δy mentioned in Theorem 2. The regularization is therefore applied only where it is needed while the structure-exploiting complexity of the Riccati recursion is preserved.

3.3 Improved stage regularization strategy

Section 3.2 fixes when and where the regularization is applied, but leaves open how the block adjustment E_k is computed. In this section, we will analyze the case where the common-used minimum perturbation strategy in regularization fails in stage regularization, which causes a severe condition number issue in the preceding stages due to the propagation of the negative curvature through the stages. This motivates us to propose an improved stage regularization strategy that avoids minimum perturbation and takes the propagation into account.

3.3.1 Failure of minimum-perturbation regularization

In this subsection, we will show that the minimum perturbation strategy, i.e. the PROJECT, which is commonly the first choice for regularization with the least modification of the original problem, can cause condition number issue in the stage regularization by introducing a general example.

Suppose at the stage k, the reduced Hessian block $\tilde{M}_k$ is not positive definite. Without loss of generality, we can assume $\tilde{M}_k$ has only one negative eigenvalue λ_j, and the corresponding eigenvector is v_j. The minimum perturbation strategy is applied as:

$$E_k = (-\lambda_j + \epsilon) v_j v_j^\top, \tag{3.6}$$

where the ϵ is a small positive value to ensure the positive definiteness of the regularized reduced Hessian block.

By Sherman-Morrison formula, the inverse of the regularized reduced Hessian block can be expressed as:

$$\begin{aligned}(\tilde{M}_k + E_k)^{-1} &= \tilde{M}_k^{-1} + \frac{\lambda_j - \epsilon}{\epsilon \lambda_j} v_j v_j^\top \\ &= \tilde{M}_k^{-1} + (\frac{1}{\epsilon} - \frac{1}{\lambda_j}) v_j v_j^\top,\end{aligned}$$

and the update of cost-to-go matrix P_k follows as:

$$\begin{aligned}P_k &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top K_k \\ &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k - (\frac{1}{\epsilon} - \frac{1}{\lambda_j}) \Gamma_k^\top v_j v_j^\top \Gamma_k \\ &= P_k^{\text{unreg}} - (\frac{1}{\epsilon} - \frac{1}{\lambda_j}) \Gamma_k^\top v_j v_j^\top \Gamma_k,\end{aligned} \tag{3.7}$$

note that this negative rank-1 modification can already render P_k itself indefinite, independently of its downstream effect on $\tilde{M}_{k-1}$, whenever the perturbation magnitude exceeds the smallest eigenvalue of P_k^{unreg}. This can happen even when the perturbation on $\tilde{M}_k$ is moderate, provided that the negative-curvature direction v_j is strongly coupled to the state through Γ_k.

Consequently, the reduced Hessian block $\tilde{M}_{k-1}$ at the previous stage as:

$$\tilde{M}_{k-1} = \tilde{M}_{k-1}^{\text{unreg}} - (\frac{1}{\epsilon} - \frac{1}{\lambda_j}) z_j z_j^\top, \tag{3.8}$$

where $z_j = B_{k-1}^\top \Gamma_k^\top v_j$ is a column vector.

And the condition number of the regularized reduced Hessian block $\tilde{M}_{k-1}$ can be approximated as:

$$\kappa(\tilde{M}_{k-1}) \approx \frac{c(\epsilon + |\lambda_j|)}{\epsilon|\lambda_j|}, \tag{3.9}$$

where $c = \frac{\|z_j\|^2}{\lambda_{\min}(\tilde{M}_{k-1}^{\text{unreg}})}$ as an $O(1)$ constant.

Notably, as the ϵ is a small positive value, the presence of the negative rank-1 update $-\frac{1}{\epsilon} z_j z_j^\top$ will dominate, not only severely deteriorates its condition number, but also likely to make $\tilde{M}_{k-1}$ indefinite, which can further cause a domino effect through the stages.

3.3.2 Insights and improved stage regularization

This issue reveals two insights in stage regularization:

- The minimum modification objective [23] is not the foremost consideration for stage regularization, as it can cause severe condition number issue in the preceding stages. A trade-off between perturbation norm and condition number must be carefully treated.
- The cost-to-go matrix P_k must be incorporated into the regularization design, as the severity of propagation depends on the coupling structure between stages, which a purely local scheme cannot capture.

From the first insight, instead of the minimum perturbation strategy, an intuitive, but effective, approach is to use the absolute value of the negative eigenvalues or inertia as the perturbation magnitude, which can avoid the severe condition number issue in the preceding stages. When it comes to the eigenvalue modification strategy, this is the MIRROR strategy, and when it comes to the modified Cholesky factorization strategy, this is the Type-II strategy [23], of which GMW is a common representative.

From the second insight, we can incorporate the cost-to-go matrix P_k into the strategy design, which also can be achieved by adding an adjustment matrix $E_{k+1,P}^i$ to the cost-to-go matrix P_k directly, so that the propagation of the negative curvature can be taken into account in the regularization design.

The resulting improved stage regularization strategy can be summarized as follows:

Algorithm 5 Backward Pass of Improved Modified Riccati Recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\text{stat},x}$, $\mathcal{E}(\cdot)$: MIRROR or Type-II strategy

for $k = N-1$ **down to** 0 **do**
  $P^{\text{reg}}_{k+1} \leftarrow P_{k+1}$ ▷ Default with no regularization
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P^{\text{reg}}_{k+1} B_k$
  $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  if factorization fails **then**
    $P^{\text{reg}}_{k+1} \leftarrow \mathcal{E}(P_{k+1})$ ▷ Regularize cost-to-go matrix
    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P^{\text{reg}}_{k+1} B_k$
    $\tilde{M}^{\text{reg}}_k \leftarrow \mathcal{E}(\tilde{M}_k)$ ▷ Regularize reduced Hessian block
    $L_k \leftarrow \text{Cholesky}(\tilde{M}^{\text{reg}}_k)$
  end if
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P^{\text{reg}}_{k+1} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}^k_{\text{stat},u} + B_k^\top (P^{\text{reg}}_{k+1} r^k_{\text{eq}} + p_{k+1}) \right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P^{\text{reg}}_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}^k_{\text{stat},x} + A_k^\top (P^{\text{reg}}_{k+1} r^k_{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

In Algorithm 5, when the Cholesky factorization fails, the regularization acts at two levels. Applying $\mathcal{E}(\cdot)$ to the cost-to-go matrix proactively convexifies the term that enters the current stage, so that the negative-curvature propagation identified in the second insight is accounted for in $\tilde{M}_k$, Γ_k, and P_k consistently. Applying it again to the reduced Hessian block acts as a local safeguard, ensuring that the Cholesky factorization succeeds. Both corrections are produced by the same non-minimal strategy $\mathcal{E}(\cdot)$, MIRROR or Type-II, rather than by the minimum-modification rule, which keeps the condition number of the regularized blocks bounded, as motivated by the first insight. When no factorization failure occurs, the recursion reduces to the standard Riccati backward pass.

3.4 Asymptotic inactivity of the regularization

Since the regularization is applied only when the detection in Section 3.2.1 fires, its behavior close to the solution is governed by the reduced Hessian $\tilde{M}_i$. In this section, we will prove that any regularization triggered by this detection is asymptotically inactive, which means that the regularization will be inactive when

the iterates are sufficiently close to the optimal solution (y^*, λ^*), with inspiration from [24], but focusing on the reduced Hessian $\tilde{M}_i$.

We consider the regularized KKT formulation, and apply the null space method to (2.6) first. The resulting formulation is given as:

$$\begin{bmatrix} M + E_i & Z^\top G^\top \\ GZ & -\Lambda_i^{-1} T_i \end{bmatrix} \begin{bmatrix} \Delta y_z \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} r^i_{\text{stat,redu}} \\ \tilde{r}^i_{\text{ineq,redu}} \end{bmatrix}, \tag{3.10}$$

where

$$\begin{aligned} M &= Z^\top H Z + E_i \\ r^i_{\text{stat,redu}} &= Z^\top r^i_{\text{stat}} + Z^\top H Y \Delta y_y, \\ \tilde{r}^i_{\text{ineq,redu}} &= r^i_{\text{ineq}} + GY \Delta y_y - \Lambda_i^{-1} r^i_{\text{comp}}. \end{aligned}$$

This formulation eliminates the equality constraints, recasting the optimal control structure into a general QP form, and yields a saddle-point system coupling the reduced primal variable Δy_z and the dual variable $\Delta\lambda$. Eliminating $\Delta\lambda$ by a further Schur complement recovers the complemented reduced Hessian of (2.9) as:

$$M + Z^\top G^\top T_i^{-1} \Lambda_i G Z + E_i = \tilde{M}_i + E_i.$$

We will make the following assumptions for the proof:

Assumption 1. *(y^*, λ^*) is a local (or global) minimizer for (2.2).*

Assumption 2. *Each local minimizer (y^*, λ^*) for (2.2) is isolated.*

Assumption 3. *Linear independence constraint qualification (LICQ) holds at the minimizer (y^*, λ^*).*

Assumption 4. *Strict complementarity holds at the minimizer (y^*, λ^*).*

Assumptions 1–3 imply the second order sufficient condition (SOSC) for local optimality, so we have the following Lemma 3:

Lemma 3. *Under Assumptions 1–3, for all $v \in C(y^*, \lambda^*) \setminus \{0\}$, it holds that $v^\top H v > 0$.*

Proof. With Assumption 1, the second order necessary condition (SONC) for local optimality holds, which states that for all $v \in C(y^*, \lambda^*)$, we have $v^\top H v \geq 0$.

We proceed by contradiction. Suppose there exists $v \in C(y^*, \lambda^*) \setminus \{0\}$ such that $v^\top H v = 0$. Consider the point $y(\alpha) = y^* + \alpha v$ for $\alpha > 0$. Because the constraints are linear and v is in the critical cone, $y(\alpha)$ remains feasible for sufficiently small α.

Since the objective function of (2.2) is quadratic, expanding $f(y(\alpha))$ yields:

$$f(y(\alpha)) = f(y^*) + \alpha \nabla f(y^*)^\top v + \frac{1}{2}\alpha^2 v^\top H v$$

As $\nabla f(y^*)^\top v = 0$, given the assumption that $v^\top H v = 0$, the equation reduces exactly to $f(y(\alpha)) = f(y^*)$. This implies that $y(\alpha)$ is also a local minimizer, which directly contradicts Assumption 2. Therefore, $v^\top H v > 0$ must hold. □

Proposition 4. *Under Assumptions 1–4, let the regularization be triggered by the Riccati-based detection. As* $\{(y_i, \lambda_i)\} \to (y^*, \lambda^*)$, $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$ *and hence* $E_i = 0$ *for all* i *sufficiently large.*

Proof. Under Assumption 4, the critical cone $C(y^*, \lambda^*)$ reduces to the tangent space of the active constraints. Therefore, we have $C(y^*, \lambda^*) = \{v \mid Fv = 0,\ G_{\mathcal{A}} v = 0\}$.

For any v with $\|v\| = 1$, write $d = Zv \in \{d \mid Fd = 0\} \setminus \{0\}$ and let G_j denote the j-th row of G. The corresponding quadratic form of the reduced Hessian reads:

$$\begin{aligned} v^\top \tilde{M}_i v &= d^\top (H + G^\top T_i^{-1} \Lambda_i G) d \\ &= d^\top H d + \sum_j \frac{\lambda_j^i}{t_j^i} (G_j d)^2 \\ &= d^\top H d + \sum_{j \in \mathcal{A}} \frac{\lambda_j^i}{t_j^i} (G_j d)^2 + \sum_{j \notin \mathcal{A}} \frac{\lambda_j^i}{t_j^i} (G_j d)^2 \end{aligned}$$

We now proceed by contradiction. Assume that $\tilde{M}_i$ is not positive definite for infinitely many i. Then, passing to a subsequence, there exist v_i with $\|v_i\| = 1$ and $v_i \to \bar{v}$ such that $v_i^\top \tilde{M}_i v_i \leq 0$ for all i in this subsequence. Writing $d_i = Z v_i$ and $\bar{d} = Z\bar{v}$, so that $d_i \to \bar{d}$, we can distinguish two cases:

For $G_{\mathcal{A}} \bar{d} = 0$, the limit direction satisfies $\bar{d} \in C(y^*, \lambda^*) \setminus \{0\}$, so $\bar{d}^\top H \bar{d} > 0$ follows by Lemma 3. Since the remaining terms are non-negative, $v_i^\top \tilde{M}_i v_i \geq d_i^\top H d_i$ with $d_i^\top H d_i \to \bar{d}^\top H \bar{d} > 0$. Hence $v_i^\top \tilde{M}_i v_i > 0$ for all sufficiently large i, contradicting the assumption.

For $G_{\mathcal{A}}\bar{d} \neq 0$, Assumption 4 gives $\lambda_j^i \to \lambda_j^* > 0$ and $t_j^i \to 0$ for $j \in \mathcal{A}$, so the term $\sum_{j\in\mathcal{A}} \frac{\lambda_j^i}{t_j^i}(G_j d_i)^2 \to \infty$, which dominates the other two terms, and thus $v_i^\top \tilde{M}_i v_i \to \infty$, contradicting the assumption.

Therefore, we can conclude that $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$ for all i sufficiently large. By Section 3.2.1, all Cholesky factorizations in the backward pass then succeed and $E_i = 0$. □

Proposition 4 shows the regularization triggered by the Riccati-based detection is asymptotically inactive: once the iterates are close to (y^*, λ^*), $E_i = 0$ and the regularized recursion coincides with the unregularized one. The solver thus recovers the exact Newton step and inherits the local convergence rate of the underlying interior-point method, without the regularization affecting its asymptotic behavior.

Chapter 4

A benchmark for indefinite OCP quadratic programs

In order to evaluate the performance of the proposed Riccati-based primal regularization, especially in the context of the subproblems of SQP in NMPC. In this chapter, we aim at constructing a benchmark of indefinite OCP QPs that arise in the SQP iterations of NMPC problems. We will first introduce the construction of the QP benchmark, including the formulation of QPs in benchmark, the OCP test problems and the strategy for collecting indefinite QPs in Section 4.1. Then we will analyze the collected problem instances to gain a better understanding of characteristics regarding definiteness in the indefinite OCP QP benchmark in Section 4.2.

4.1 Benchmark construction

In this section, we will introduce the construction of the indefinite QP benchmark. We will first introduce the formulation of QPs in benchmark, which originate from the SQP iterations of the OCPs, and then we will introduce the eight OCP test problems used to construct the benchmark, which are formulated as fixed-time regulator problems with quadratic stage costs and box constraints on the controls. Then the collection strategy is designed to ensure that the collected QPs are representative of the indefinite QPs that arise in practice, and that they share the mathematical structure of the OCPs.

4.1.1 Formulation of QP subproblems from OCPs

We will start from the formulation of the OCPs to obtain the resulting QP subproblems that arise in the SQP iterations. In the benchmark, all optimal control problems share the same structure: a fixed-horizon, direct multiple-shooting [1] optimal control problem with a quadratic stage cost that penalizes deviations from given state and control references. The mathematical formulation of the OCPs is given as follows:

$$\begin{aligned}
\underset{x,\,u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \frac{1}{2} \begin{bmatrix} x_k - x_k^{\text{ref}} \\ u_k - u_k^{\text{ref}} \end{bmatrix}^\top \begin{bmatrix} \mathcal{Q} & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R} \end{bmatrix} \begin{bmatrix} x_k - x_k^{\text{ref}} \\ u_k - u_k^{\text{ref}} \end{bmatrix} \\
& + \tfrac{1}{2} \|x_N - x_N^{\text{ref}}\|_{\mathcal{Q}_N}^2 \\
\text{subject to} \quad & x_0 = \bar{x}_0, \\
& x_{k+1} = f_k(x_k, u_k), \qquad k = 0, \ldots, N-1, \\
& \underline{g} \leq \mathcal{C}_k x_k + \mathcal{D}_k u_k \leq \bar{g}, k = 0, \ldots, N-1, \\
& \underline{h} \leq h_k(x_k, u_k) \leq \bar{h}, \quad k = 0, \ldots, N-1, \\
& \underline{g}_N \leq \mathcal{C}_N x_N \leq \bar{g}_N,
\end{aligned} \tag{4.1}$$

where $x_{k+1} = f_k(x_k, u_k)$ is one integration step of the continuous dynamics over $[t_k, t_{k+1}]$ for notational simplicity, $\bar{x}_0$ is the given initial state, $\mathcal{Q}, \mathcal{R}, \mathcal{Q}_N \succeq 0$ are weighting matrices, $\mathcal{S}_k$ is the state–control cost-coupling matrix, and $x_k^{\text{ref}}, u_k^{\text{ref}}$ are the reference points. $\mathcal{C}x_k + \mathcal{D}u_k$ includes the affine and box path constraints, h_k collects the nonlinear path constraints, and $\underline{g}_N \leq \mathcal{C}_N x_N \leq \bar{g}_N$ are the terminal inequality constraints.

When using the exact Hessian of the Lagrangian in the SQP iterations, the resulting Hessian of the QP subproblem at stage $k = 0, \ldots, N-1$ reads:

$$\begin{aligned}
H_k &= \nabla^2 \mathcal{L}(x, u, \pi, \lambda) \\
&= \begin{bmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{bmatrix} + \sum_{j=1}^{n_x} (\pi_k)_j \nabla^2 [f_k]_j(x_k, u_k) + \sum_{j=1}^{n_c} (\lambda_k)_j \nabla^2 [h_k]_j(x_k, u_k),
\end{aligned} \tag{4.2}$$

where n_x is the dimension of states and n_c is the number of nonlinear constraints, π_k and λ_k are the Lagrange multipliers associated with the dynamics and nonlinear constraints, respectively.

The stage cost is quadratic, so its Hessian $\left[\begin{smallmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{smallmatrix}\right]$ is constant and exact. The indefiniteness of H_k thus has two sources: a constant contribution from the

cost block, and an iterate-dependent contribution from the two curvature terms, which is usually the dominant one during the SQP iterations. In these curvature terms, each component Hessian $\nabla^2[f_k]_j$ and $\nabla^2[h_k]_j$ is symmetric but carries no definiteness guarantee, and the associated multipliers $(\pi_k)_j$ and $(\lambda_k)_j$ may take arbitrary sign and magnitude during the SQP iterations. Their weighted sum is thus a general symmetric and typically indefinite matrix. Adding it to the constant cost block can introduce negative eigenvalues in H_k, in particular when the multipliers are large, i.e. when constraints are strongly active or the current iterate is still far from a solution. Consequently, the exact-Hessian QP subproblem is in general nonconvex.

Importantly, this indefiniteness is confined to the individual stage blocks: H_k remains block-diagonal across stages and, together with the banded Jacobian induced by the shooting constraints, preserves the characteristic sparse structure of the OCP.

Collecting the stage Hessians H_k and Jacobians and linearizing the constraints of (4.1) at the current SQP iterate yields a structured QP. The subproblem takes the form:

$$
\begin{aligned}
\underset{\Delta x,\,\Delta u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \left(\tfrac{1}{2} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix}^\top \begin{bmatrix} Q_k & S_k^\top \\ S_k & R_k \end{bmatrix} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix} + \begin{bmatrix} q_k \\ r_k \end{bmatrix}^\top \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix} \right) \\
& + \tfrac{1}{2}\, \Delta x_N^\top Q_N\, \Delta x_N + q_N^\top \Delta x_N \\
\text{subject to} \quad & \Delta x_0 = \Delta \bar{x}_0, \\
& \Delta x_{k+1} = A_k \Delta x_k + B_k \Delta u_k + c_k, \qquad k = 0, \dots, N-1, \\
& \begin{bmatrix} \underline{u}_k \\ \underline{x}_k \\ \underline{d}_k \end{bmatrix} \leq \begin{bmatrix} J_k^{b,u} & 0 \\ 0 & J_k^{b,x} \\ D_k & C_k \end{bmatrix} \begin{bmatrix} \Delta u_k \\ \Delta x_k \end{bmatrix} \leq \begin{bmatrix} \overline{u}_k \\ \overline{x}_k \\ \overline{d}_k \end{bmatrix}, k = 0, \dots, N-1, \\
& \underline{\Delta x}_N \leq J_N^{b,x} \Delta x_N \leq \overline{\Delta x}_N.
\end{aligned}
\tag{4.3}
$$

where, at each stage k, $\Delta x_k \in \mathbb{R}^{n_x}$ and $\Delta u_k \in \mathbb{R}^{n_u}$ denote the state and control deviation variables, and N is the horizon length.

All remaining quantities are obtained by linearizing the OCP at the current SQP iterate (x_k, u_k) with multipliers (π_k, λ_k):

- Q_k, R_k, and S_k are the state, control, and state–control coupling blocks of the exact Lagrangian Hessian H_k. $q_k = \mathcal{Q}_k(x_k - x_k^{\text{ref}}) + \mathcal{S}_k^\top(u_k - u_k^{\text{ref}})$ and $r_k = \mathcal{R}_k(u_k - u_k^{\text{ref}}) + \mathcal{S}_k(x_k - x_k^{\text{ref}})$ are the corresponding stage-cost gradients,

and Q_N, q_N are the terminal Hessian and gradient.

- $A_k = \nabla_{x_k} f_k$ and $B_k = \nabla_{u_k} f_k$ are the Jacobians of the discretized dynamics, and $c_k = f_k(x_k, u_k) - x_{k+1}$ is the dynamics (shooting) residual at the current iterate; together they define the linearized state-transition equation. At a dynamically feasible iterate $c_k = 0$.
- $\Delta\bar{x}_0 = \bar{x}_0 - x_0$ is the residual of the initial equality constraint, i.e. the correction needed to satisfy $x_0 = \bar{x}_0$ after the step.
- $J_k^{b,u}$ and $J_k^{b,x}$ are selection matrices picking out the bounded control and state components, with bounds $\underline{u}_k, \overline{u}_k$ and $\underline{x}_k, \overline{x}_k$ (box constraints).
- $C_k = \nabla_{x_k} h_k$ and $D_k = \nabla_{u_k} h_k$ are the Jacobians of the general path constraints, with $\underline{d}_k = \underline{h} - h_k(x_k, u_k)$ and $\overline{d}_k = \overline{h} - h_k(x_k, u_k)$; these constraints couple states and controls. For the affine constraints $\mathcal{C}x_k + \mathcal{D}u_k$ of (4.1), C_k and D_k reduce to the constant matrices $\mathcal{C}$ and $\mathcal{D}$.

4.1.2 Benchmark OCPs

In this subsection, we introduce the formulation of the eight OCPs used to construct the benchmark. Each OCP is obtained by taking a dynamic system provided in [25] and casting it as a regulation or point-to-point transfer problems with problem-specific modifications, which can be described as (4.1). The eight OCPs are named and characterized as follows:

- **Pendulum:** An inverted pendulum on an actuated cart, driven by a horizontal force. It is regulated from the downward to the upright equilibrium subject to an input bound and an additional mixed state–input inequality constraint. Its stage cost further includes a bilinear state–input term, which facilitates the occurrence of indefinite Hessians in the SQP iterations.
- **Acrobot:** Adapted from [26]. A two-link pendulum actuated only at the elbow joint, with joint friction. The state is driven between two prescribed configurations through terminal equality constraints, yielding a strongly nonlinear, underactuated maneuver.
- **Dubins car:** Adapted from [27]. A kinematic unicycle with position and heading as states, actuated by forward velocity and turn rate. It performs a parallel-parking maneuver, a lateral displacement returning to the initial heading, within a narrow corridor imposed by state-box constraints, with a terminal equality constraint on the full pose.
- **Furuta:** Adapted from [28]. A rotary inverted pendulum with experimen-

tally identified parameters and joint damping, actuated by a single base-arm torque. The pendulum is swung up and stabilized at the upright position under tight torque limits and a non-uniform time grid.

- **Flying robot:** Adapted from [29]. A planar free-flying robot with position, orientation and their rates, propelled by four non-negative thrusters that together produce a body-aligned net force and a turning moment. The state is driven toward a reference point—zero position and orientation with a prescribed nonzero translational velocity—under thrust non-negativity and summed-thrust linear constraints.
- **Robot arm:** Adapted from [29]. A three-DOF robot arm (radial extension, rotation, elevation) with configuration-dependent inertia and a cross-axis actuator coupling that is nonlinear in the inputs. A joint angle is regulated to a new set-point under state and input bounds; the input coupling renders the exact Hessian indefinite.
- **Crane:** Adapted from [29]. An overhead crane with hoisting length, trolley position and payload swing angle, actuated by trolley and hoisting accelerations. The load is transferred between two rest positions while respecting a swing-rate bound and a nonlinear region constraint on the payload.
- **Moonlander:** Adapted from [30]. A planar lunar lander with translational and rotational motion under reduced gravity, actuated by two laterally offset thrusters. It performs a soft landing, reaching a ground target with zero terminal velocity, under non-negative, bounded thrust.

Table 4.1: Summary of the evaluated OCP problems. The state dimension n_x, control dimension n_u, and horizon length N span a range of common control problem scales.

Problem	n_x	n_u	N
Pendulum	4	1	25
Acrobot	4	1	15
Dubins car	3	2	25
Furuta	4	1	20
Flying robot	6	4	50
Robot arm	6	3	30
Crane	6	2	40
Moonlander	6	2	30

Besides the different physical systems, These OCPs also cover different state

and control dimensions, as shown in Table 4.1. The state dimension n_x ranges from 3 to 6, the control dimension n_u from 1 to 4, and the horizon length N from 15 to 50. This spread is chosen so that the benchmark spans a range of common problem scales in control problems, which allows the evaluated methods to be assessed across problems of varying size and structure.

4.1.3 Collecting strategy

Based on the formulated OCPs, we can construct a representative benchmark of indefinite QPs arising from SQP-based NMPC, avoiding the artificial assumptions of randomly generated Hessians and constraints. Specifically, we propose a collection strategy that extracts the QP subproblems directly from intermediate SQP iterations. By logging the linearized dynamics, stage constraints, and indefinite objective approximations exactly as they arise in each iteration, this procedure ensures that every benchmark instance faithfully reflects the numerical conditioning and sparsity encountered in practice.

The collection strategy consists of two steps: *generate* and *collect*, and can be summarized as follows:

1. **Generate:** In this step, we generate iterates by solving the OCP. To obtain iterates at various points during the solution process, we can apply SQP-level regularization techniques (such as PROJECT or MIRROR) or utilize the Gauss-Newton Hessian approximation in `acados`.
2. **Collect:** In this step, we utilize the previously generated iterates as initial guesses for linearization within the solver. By restricting the maximum number of SQP iterations, we can selectively collect indefinite QPs that exhibit an indefinite reduced Hessian.

In the collect step, QPs are all with one failing status code from `acados`, code `4` for infeasible QP, which indicating the indefinite reduced Hessian causing the failing of the Riccati recursion in `HPIPM`. This is achieved by setting the Riccati algorithm in HPIPM as classical Riccati recursion, i.e. `ric_alg = 0`, instead of the square-root version [31], which requires the reduced Hessian to be positive definite in order to proceed with the Riccati recursion. So when the HPIPM fails to proceed with the Riccati recursion, it indicates that the reduced Hessian is indefinite, and the corresponding QP is collected.

For each problem, we limit the number of near-identical QPs, where similarity is measured by the minimal eigenvalue and the rate of indefinite stages (both

introduced in Section 4.2), so that the number of instances per problem stays within a comparable range and the collected instances remain meaningful. We further use IPOPT [20] as a reference solver and keep only the QPs that it solves successfully. After this filtering, we obtain a total of 175 indefinite QPs from the eight OCPs, the per-problem counts are reported in Table 4.2 and range from 15 to 29.

Table 4.2: Number of indefinite QPs collected from each OCP (175 in total).

Problem	**Number**
Pendulum	18
Acrobot	15
Dubins car	29
Furuta	19
Flying robot	20
Robot arm	24
Crane	29
Moonlander	21

The QPs are stored in `json` format and are available at `https://github.com/Pandatheon/Primal-dual_IPM.git`.

4.2 Benchmark analysis

In this section, we will introduce some analysis of the collected problem instances for gaining a better understanding of characteristics regarding definiteness in the indefinite OCP QP benchmark, The distribution of indefiniteness is analyzed in Section 4.2.1, which describes the magnitude and coverage of the indefinite reduced Hessians in the collected QPs, and the source of indefiniteness is analyzed in Section 4.2.2, which further illustrates the source of indefiniteness in the collected QPs by analyzing the eigenvalues of the stage Hessians.

4.2.1 Distribution of indefiniteness

We first characterize the collected QPs along two axes: the *depth* of indefiniteness, measured by the magnitude of the smallest (most negative) eigenvalue $|\lambda_{\min}|$ of the reduced Hessian, and the *width*, measured by the fraction of block reduced Hessians that are indefinite within a QP (the indefinite-stage rate).

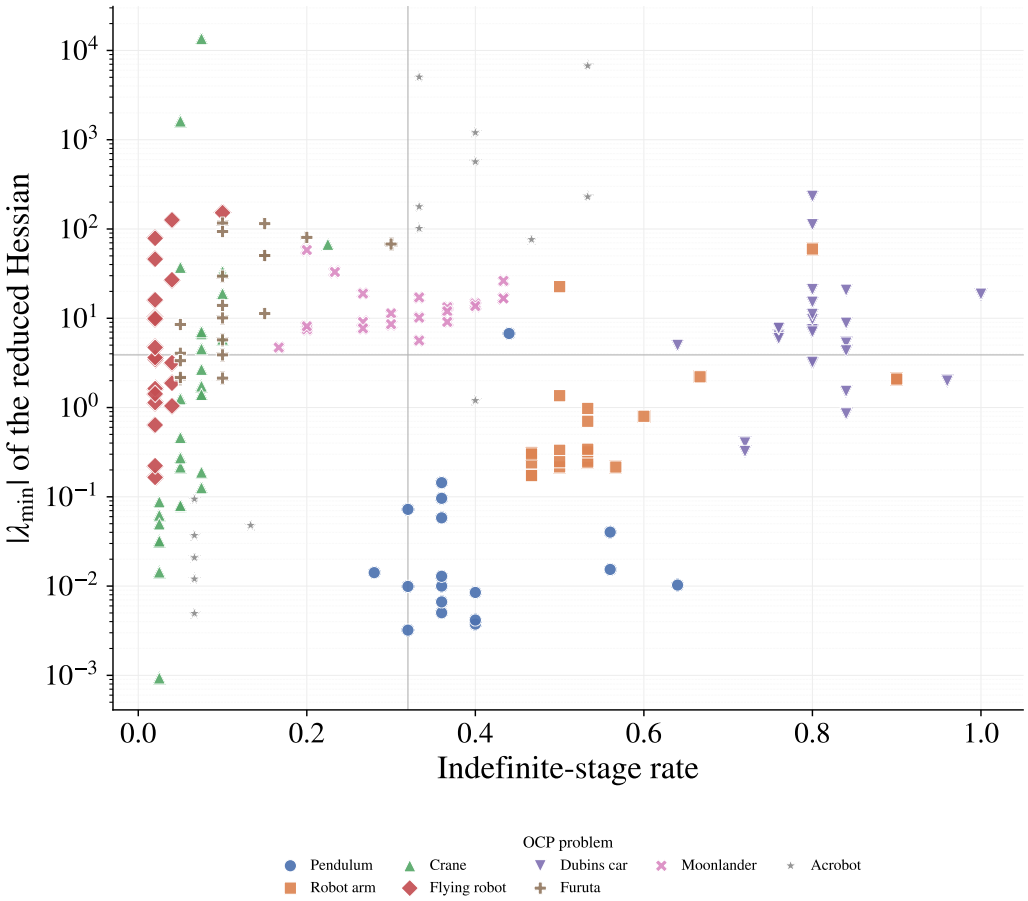
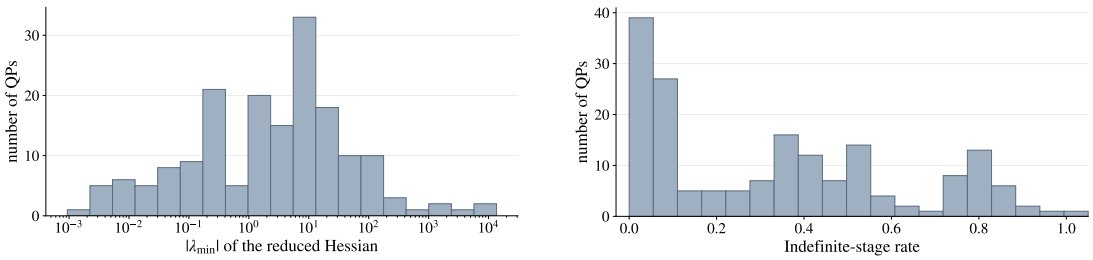


Figure 4.1: Depth ($|\lambda_{\min}|$ of the reduced Hessian, log scale) versus width (indefinite-stage rate) of the collected QPs, coloured by OCP problem.

As seen from Figure 4.1, the depth of the collected QPs ranges from near-singular to very large negative curvature, and each OCP occupies its own region of the depth–width plane. Problems that are indefinite in only a small fraction of their stages, such as *Crane*, *Flying robot* and *Furuta* (indefinite-stage rates below 0.30), can still reach deep negative curvature. Conversely, *Dubins car* and *Robot arm*, with indefinite-stage rates from 0.47 up to 1, show only moderate depth. In the collected QPs, wide and deep indefiniteness thus seldom occur together.

Moonlander and *Acrobot* lie in between, with intermediate indefinite-stage rates. *Moonlander* stays within a narrow depth band, whereas *Acrobot* spans almost the full depth range of the benchmark. *Pendulum* contributes QPs with very shallow negative curvature at intermediate rates. The eight OCPs thus cover narrow-but-deep, intermediate and wide-but-moderate indefiniteness.



(a) Depth: distribution of $|\lambda_{\min}|$.

(b) Width: distribution of the indefinite-stage rate.

Figure 4.2: Marginal distributions of depth and width over all collected QPs.

Figure 4.2 shows the marginal distributions of the same data. The depth histogram (Figure 4.2a) peaks around $|\lambda_{\min}| \approx 10$ and thins out toward both ends, with only five QPs in the very deep tail. The negative curvature encountered during the SQP iterations is therefore typically of moderate magnitude. The width histogram (Figure 4.2b) covers the full range up to 1, with a large share of QPs indefinite in only a few stages: close to 40% have a rate below 0.11. The remaining QPs form smaller groups at intermediate and high rates, and only one QP is indefinite in essentially all of its stages.

4.2.2 Source of indefiniteness

Recall that the block reduced Hessian at stage k, $M_k = R_k + B^\top P_{k+1} B$, is assembled from two contributions: the control cost block R and the cost-to-go block P.

To understand where the indefiniteness originates, we classify every indefinite stage according to which of the two blocks has a negative smallest eigenvalue: R *only* (R indefinite, P definite), P *only* (P indefinite, R definite), and *both* (both indefinite).

Figure 4.3 aggregates all indefinite stages across the benchmark. In 1114 of the 1571 indefinite stages (71%), *both* R and P are indefinite. Of the remaining stages, 299 (19%) are P *only* and 158 (10%) are R *only*. In total, P is indefinite in 1413 stages (90%) and R in 1272 (81%).

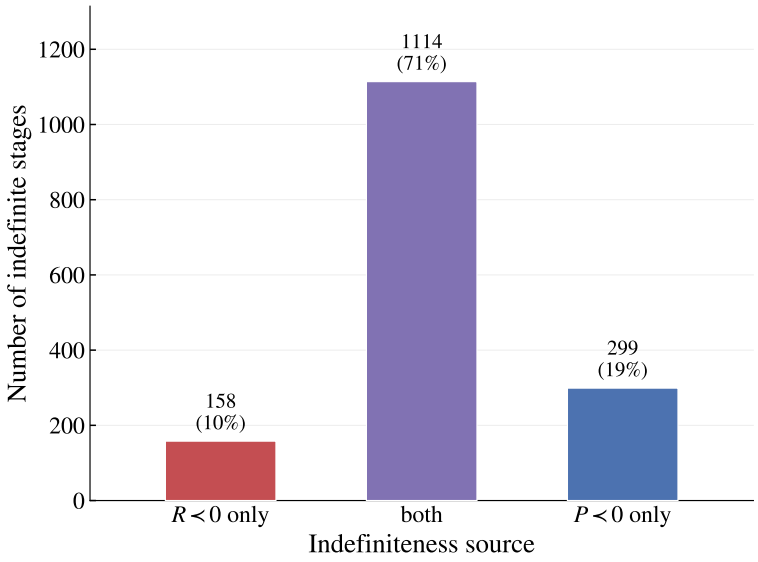


Figure 4.3: Number of indefinite stages by source: control cost R, cost-to-go P, or both.

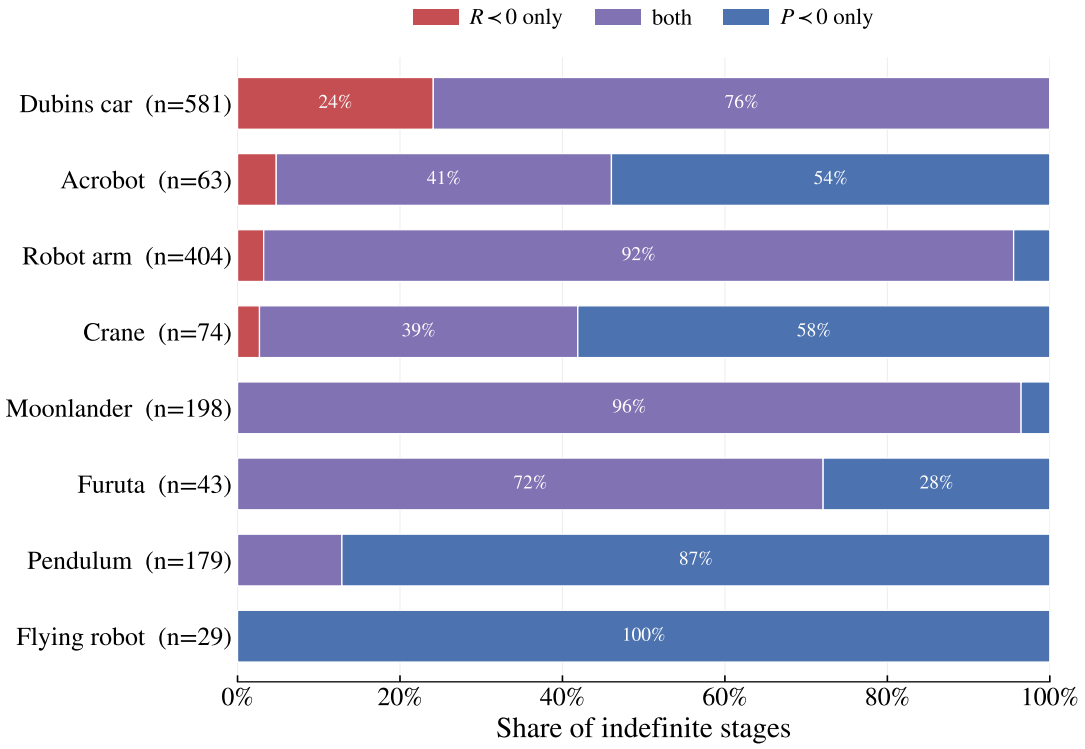


Figure 4.4: Per-problem composition of stage indefiniteness. Each bar shows, for one OCP, the share of stages that are R *only*, *both*, P *only*, (n is the number of collected indefinite stages).

Figure 4.4 breaks the same classification down per problem. Six of the eight OCPs are dominated by a single category ($\geq 70\%$ of their stages), and only *Acrobot* and *Crane* are mixed. The *both* case dominates *Moonlander* (96%), *Robot arm* (92%), *Dubins car* (76%) and *Furuta* (72%). *Dubins car* ($n = 581$), *Robot arm* ($n = 404$) and *Moonlander* ($n = 198$) also contribute the most indefinite stages, so they account for most of the aggregate *both* share. The *P only* case dominates *Flying robot* (100%) and *Pendulum* (87%), and is also the largest category in the two mixed problems, *Crane* (58% against 39% *both*) and *Acrobot* (54% against 41% *both*). The *R only* case appears in four problems and is substantial only for *Dubins car* (24% of its stages, i.e. 140 of the 158 *R only* stages in the benchmark); in *Robot arm*, *Acrobot* and *Crane* it stays below 5%.

In conclusion, the cost-to-go block is involved in 90% of the indefinite stages, so a regularization strategy that merely enforces $M \succ 0$ would leave the negative curvature unaddressed in most stages. Since the cost-to-go term enters the reduced Hessian through $M_k = R_k + B^\top P_{k+1} B$, a regularization method must keep P_{k+1} (and hence M_k) definite; treating the control cost block alone is not sufficient.

Chapter 5

Numerical experiments

In this chapter, we evaluate the proposed regularization methods numerically and discuss the results. We begin in Section 5.1 with the two metrics used throughout the comparison, the iteration profile and the flop unit estimation, which measure convergence behavior and factorization cost respectively. Section 5.2 then describes the prototype solver on which all experiments are run and establishes that it reproduces the iteration behavior of HPIPM on convex QPs, so that any difference observed later can be attributed to the regularization rather than to the solver itself. Section 5.3 fixes the experiment setup, naming and classifying the regularization methods under comparison and stating the parameters shared by all of them. Section 5.4 presents the numerical comparison, first among the Riccati-based methods themselves and then against full Hessian regularization and inertia correction. Section 5.5 concludes the chapter by discussing what distinguishes the hard instances from the easy ones and why the full Hessian methods perform poorly.

5.1 Experiment metrics

In this section, we will introduce the metrics used to evaluate the performance of different regularization strategies in our experiments. Since we care about both the convergence behavior and the factorization cost caused by different regularization methods, we will use two complementary metrics: iteration profile and flop unit estimation. The former one, iteration profile, is used to evaluate how many iterations each method takes to converge on a set of test instances, while the latter one, flop unit estimation, is approximated by the number of modified Cholesky factorization of the reduced Hessian, which is used to evaluate the factorization cost of each method.

5.1.1 Iteration profile

Let $\mathcal{P}$ denote the set of test instances and $\mathcal{M}$ the set of candidate regularization strategies. For a method $m \in \mathcal{M}$ and an instance $\mathrm{p} \in \mathcal{P}$, let $i_{m,\mathrm{p}}$ be the number of iterations m needs to satisfy the convergence criterion on p. If the run reaches the iteration cap $i_{\max}$, the instance is counted as unsolved and $i_{m,\mathrm{p}} = \infty$. The *iteration profile* of method m is

$$\Pi_m(\ell) \;=\; \Big|\{\, \mathrm{p} \in \mathcal{P} : i_{m,\mathrm{p}} \le \ell \,\}\Big|, \qquad \ell = 1, \dots, i_{\max}, \tag{5.1}$$

i.e. the number of instances that m solves within ℓ iterations. A curve that rises steeply at small ℓ indicates an *efficient* method, and its final value $\Pi_m(i_{\max})$, the number of solved instances, measures its *robustness*.

5.1.2 Flop unit estimation

Iteration counts alone do not reflect the work performed per iteration, which differs substantially across regularization strategies: an eigenvalue-based correction and a modified-Cholesky correction may reach convergence in a comparable number of iterations while incurring very different arithmetic cost. To compare methods on a common scale, we express the cost of all matrix factorizations and decompositions in a single flop unit.

We take one modified Cholesky factorization of the block reduced Hessian as the unit. For a control dimension n_u, factorizing an $n_u \times n_u$ matrix costs approximately $\frac{1}{3}n_u^3$ flops, hence

$$\text{unit} \;\approx\; \tfrac{1}{3}\, n_u^3. \tag{5.2}$$

The modified factorization differs from the plain one only by lower-order diagonal updates, so to leading order it also costs one unit.

The remaining primitives are expressed relative to this unit. A symmetric eigenvalue decomposition of the same block costs about $9n_u^3$ flops [32], i.e.

$$\frac{9n_u^3}{\frac{1}{3}n_u^3} \;=\; 27 \text{ units}. \tag{5.3}$$

Regularizing the cost-to-go matrix $P_{k+1} \in \mathbb{R}^{n_x \times n_x}$ or a full Hessian block $\tilde{H}_k \in$

$\mathbb{R}^{(n_x+n_u)\times(n_x+n_u)}$ applies the same primitives to a larger block. With the ratios

$$\omega_P \;=\; \left(\frac{n_x}{n_u}\right)^3, \qquad \omega_H \;=\; \left(\frac{n_x+n_u}{n_u}\right)^3, \tag{5.4}$$

a modified Cholesky factorization of P_{k+1} costs ω_P units and an eigenvalue decomposition $27\omega_P$ units, and likewise ω_H and $27\omega_H$ units for $\tilde{H}_k$. Since the full Hessian regularization modifies all N blocks $\tilde{H}_k$ at once, each regularization costs $N\omega_H$ or $27N\omega_H$ units. The cost of a solve is the total number of units accumulated over all its iterations.

5.2 Prototype solver implementation

To unify the comparison, we implemented a prototype solver capable of accommodating all regularization methods under consideration. The Riccati-based primal-dual interior-point solver is implemented in Python, using HPIPM, the state-of-the-art Riccati-based IPM solver for OCP QPs, as a reference. The prototype was designed to reproduce its iteration behavior on convex QPs, where the two solvers should therefore agree. It is available at `https://github.com/Pandatheon/Primal-dual_IPM.git`.

One implementation detail deserves mention, as it departs from the plain FTB rule stated in Section 2.4. Rather than retracting the maximal step by a fixed fraction η, the primal and dual step sizes α_{prim} and α_{dual} are each obtained from their respective maximal step through the heuristic

$$\alpha = \begin{cases} \alpha_{\max}\Big[(1-\alpha_{\max})\,\kappa_\eta + \alpha_{\max}\hat{\eta}\Big], & \alpha_{\max} < 1, \\ \alpha_{\max}, & \alpha_{\max} = 1, \end{cases} \tag{5.5}$$

where $\alpha_{\max} \in (0, 1]$ denotes the maximal step length for ensuring the positivity of the slack and dual variables, $\kappa_\eta = 0.99$ and $\hat{\eta} = 1 - 10^{-7}$. The bracketed factor interpolates linearly between the conservative constant κ_η and the near-unit value $\hat{\eta}$, with $\alpha_{\max}$ itself as the interpolation weight. The separate case $\alpha_{\max} = 1$ is needed, otherwise the first branch would scale an unblocked full step by $\hat{\eta}$ and prevent it from ever being taken exactly, thus affecting the local qudratic convergence rate of IPM.

For the comparison to be meaningful, We align the prototype solver with HPIPM in terms of the algorithmic choices. We therefore define a set of HPIPM

settings, referred to in the following as the `CUSTOMIZED` mode, the values are listed in Table 5.1.

Table 5.1: Detailed settings for the defined `CUSTOMIZED` mode.

Parameter	**Value**	**Parameter**	**Value**
mu0	1e1	lam_min	1e-12
alpha_min	1e-12	t_min	1e-12
res_g_max	1e-6	tau_min	1e-12
res_b_max	1e-8	lam0_min	1e-12
res_d_max	1e-8	t0_min	1e-12
res_m_max	1e-6	warm_start	0
dual_gap_max	1e15	abs_form	0
iter_max	30	comp_dual_sol_eq	1
stat_max	30	comp_res_exit	1
pred_corr	1	comp_res_pred	1
cond_pred_corr	1	split_step	1
itref_pred_max	0	m_safe	0.5
itref_corr_max	0	var_init_scheme	0
reg_prim	0.0	t_lam_min	2
square_root_alg	0	t0_init	2
lq_fact	0	update_fact_exit	1

As one can notice, the `CUSTOMIZED` mode largely coincides with the `SPEED` mode of the original HPIPM, differing in two respects. First, the square-root algorithm is disabled (`square_root_alg`= 0), so that the Riccati recursion is carried out directly rather than on Cholesky factors. Second, the primal and dual step sizes are computed separately (`split_step`= 1) instead of sharing a single step length. In both modes no iterative refinement is applied to the predictor or corrector steps (`itref_pred_max`=`itref_corr_max`= 0) and no default primal regularization is added (`reg_prim`= 0), while Mehrotra's predictor–corrector is retained and applied conditionally (`pred_corr`=`cond_pred_corr`= 1).

To verify that the two implementations do agree, we collected 40 convex QP subproblems, five from each of the 8 OCPs in the benchmark, and compared the iteration behavior of the prototype solver against that of `CUSTOMIZED`-mode HPIPM. The resulting iteration profiles are shown in Figure 5.1.

Both solvers solve all 40 instances in the exact same number of iterations for each instance, resulting in performance profiles that coincide across the whole budget range, from 4 up to 22 iterations. On convex QPs the prototype solver therefore reproduces the iteration behavior of HPIPM, which verifies the prototype

as a reliable baseline and allows us attribute the differences observed in the experiments to the regularization rather than to the solver.

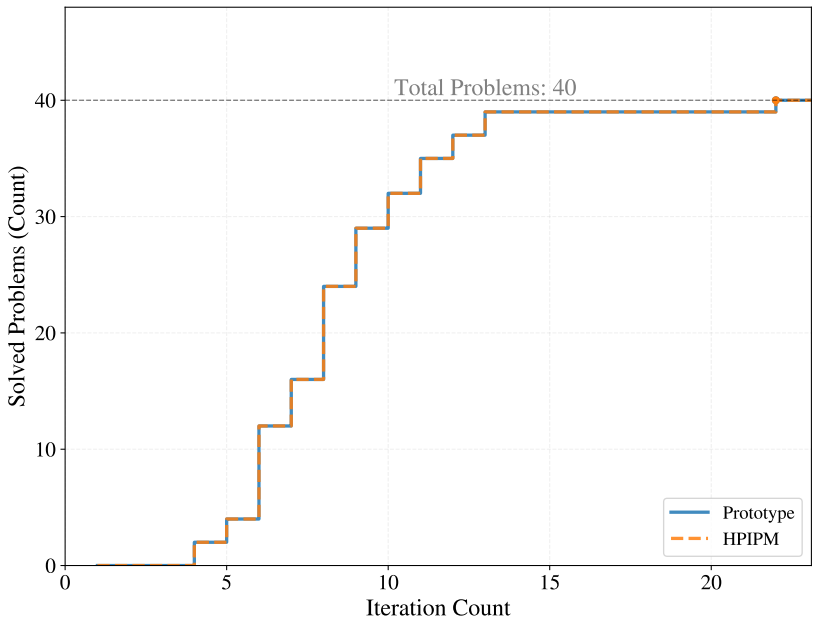


Figure 5.1: Iteration profiles on 40 convex QPs. The prototype solver is compared with HPIPM, showing identical iteration behavior.

5.3 Experiment setup

In this section, the settings preparation for the experiment are introduced. First, based on the previously discussed methods, we will name and classify the regularization methods considered in the experiment, then we will introduce the general setup of the experiments, including the convergence criterion and step-size control strategy.

5.3.1 Experimented regularizations

Before turning to the results, we fix the set of regularization methods entering the comparison, together with the naming convention used throughout this chapter.

Following the classification in Section 3.1, the methods are grouped by their *target*, that is, by the matrix whose indefiniteness the correction is derived from and to which the correction is directly applied, rather than by how the correction itself is computed. This yields three families: full Hessian regularization, Riccati-based regularization, and inertia correction. The numerical strategies

of Section 3.1.1, namely spectral projection (PROJECT), eigenvalue mirroring (MIRROR) and the Gill–Murray–Wright modified Cholesky factorization (GMW), are largely independent of the matrix they act on and can therefore be paired with each target.

Every method is accordingly named in the form TARGET_STRATEGY, so that both factors can be read off directly from the label, and the comparison can be carried out either across targets at a fixed strategy or across strategies at a fixed target.

Full Hessian Regularization The first family follows Section 3.1.3 and enforces positive definiteness of the full Hessian $\tilde{H}$ in the KKT system. The regularization is also triggered by the failure of the Choleksy factorization in Riccati recursion, the regularized full Hessian is obtained by stage-wisely regularizing each block $\tilde{H}_k$ and the Riccati recursion is re-attempted with it. The three variants differ only in how the block adjustment E_k is formed:

- H_PROJECT: the negative and near-zero eigenvalues of each $\tilde{H}_k$ are clipped to ϵ, which gives the smallest perturbation in Frobenius norm that restores positive definiteness.
- H_MIRROR: the eigenvalues of each $\tilde{H}_k$ are replaced by their absolute values (floored at ϵ), which preserves the magnitude of the curvature along every eigendirection at the price of a larger modification.
- H_GMW: the diagonal pivots of each $\tilde{H}_k$ are bounded from below during the $\mathrm{LDL}^\top$ factorization, which avoids an eigendecomposition altogether and keeps the entries of the triangular factor bounded, but yields a non-minimal, ordering-dependent perturbation.

Riccati-based Regularization The second family is the one proposed in Section 3.2 and acts on the reduced Hessian $\tilde{M}$. The algorithms are given in Algorithms 4 and 5, where the regularization is applied stage-wise to the block reduced Hessian $\tilde{M}_k$ during the backward pass, when the failure of the Cholesky factorization of $\tilde{M}_k$ occurs.

The two algorithms differ in which matrices of the backward pass the adjustment is derived from and applied to, giving two targets within this family:

- M: only the block reduced Hessian $\tilde{M}_k$ is corrected, as in Algorithm 4. The correction is purely local and serves as a minimal safeguard for the factorization.

- PM: the cost-to-go matrix P_{k+1} entering the stage is corrected as well, as in Algorithm 5 introduced in this work, so that the propagation of negative curvature analysed in Section 3.3 is accounted for consistently in $\tilde{M}_k$, Γ_k and P_k.

Pairing each target with the three strategies gives the six methods compared in this chapter. Each method accordingly answers a specific question in the comparison based on the insights mentioned in Section 3.3:

- M_PROJECT: the reference point of the family. A minimum-norm correction applied locally is the naive and cheapest way to enforce positive definiteness, and is the configuration that the analysis of Section 3.3 predicts to be the most exposed to condition-number growth.
- M_MIRROR: whether abandoning the minimum-perturbation rule is by itself sufficient, with the propagation left untreated.
- M_GMW: the same question as M_MIRROR, but with the correction on the inertia of the factorization rather than on the spectrum, which indicates whether the cheaper Type-II correction can stand in for mirroring.
- PM_PROJECT: conversely, whether regularizing the cost-to-go matrix helps even when the minimum-perturbation rule is retained, so that this case can be discussed rather than merely argued for.
- PM_MIRROR: both insights combined, and thus the method the analysis of Section 3.3 recommends.
- PM_GMW: combines both insights with a cheaper factorization-based correction to explore the trade-off between robustness and decomposition cost compared to PM_MIRROR.

Inertia Correction The third family is the Levenberg–Marquardt style inertia correction used in IPOPT [20], which we denote by INERTIA_LM and take as the practical reference point for the comparison.

In this strategy, a scalar multiple of the identity $\delta_w I$ is added stage-wise to the full Hessian, i.e. each full Hessian block $\tilde{H}_k$ is replaced by $\tilde{H}_k + \delta_w I$, and the Riccati recursion is repeated with increasing δ_w until it succeeds. Only the primal regularization from IPOPT is retained, as dual regularization is not the primary focus of this work.

Notably, the method sits between the two preceding families, as it has no distinct target matrix in the above sense. Being a uniform multiple of the identity, the shift is not tied to any particular direction, so the perturbation remains a

uniform shift along every direction for both the full and the reduced Hessian. Accordingly, as $\delta_w I$ is applied uniformly to Q_k and R_k in full Hessian blocks $\tilde{H}_k$, it enters the Riccati recursion as the same additive term on the reduced Hessian blocks $\tilde{M}_k$ and P_k.

The trial sequence follows IPOPT as well: the unmodified matrix is attempted first ($\delta_{\mathrm{LM}} = 0$), the first correction of a solve starts from δ_w^0 and is then scaled by the accelerated factor $\bar{\kappa}_w^+$, subsequent corrections are scaled by κ_w^+ until the Riccati recursion succeeds or the threshold $\delta_w^{\max}$ is exceeded, and a successful value is stored and reduced by κ_w^- as the starting point of the next iteration. The parameters are set to the IPOPT defaults and are listed in Table 5.2.

Table 5.2: Inertia Correction (IC) Parameters

Symbol	**Value**	**Description**
δ_w^0	10^{-4}	Initial Hessian perturbation parameter
$\delta_w^{\min}$	10^{-16}	Minimum allowable regularization parameter
$\delta_w^{\max}$	10^{16}	Maximum allowable regularization threshold
κ_w^-	$1/3$	Factor to decrease perturbation upon success
κ_w^+	8	Factor to increase perturbation after inertia failure
$\bar{\kappa}_w^+$	10^2	Accelerated scaling factor for severe indefiniteness
δ_{LM}	0.0	Initial LM shift (IC-1: attempt unmodified matrix first)

Taken together, the three families span ten methods, which share the same solver, the same benchmark and the same parameters introduced in the next subsection,

5.3.2 General setup

All ten methods are run under the same conditions. The barrier parameter is updated by Mehrotra probing and the step size is controlled by the FTB rule followed by the two-phase line search, while the remaining settings are collected in Table 5.3.

A run terminates either when all four residuals fall below their tolerances or when the iteration budget $i_{\max}$ is exhausted, in which case the instance counts as unsolved.

Table 5.3: Settings shared by all experiments.

Symbol	Value	Role
r_{stat}	10^{-6}	Tolerance on the stationary residual
r_{eq}	10^{-8}	Tolerance on the equality residual
r_{ineq}	10^{-8}	Tolerance on the inequality residual
r_{comp}	10^{-6}	Tolerance on the complementarity residual
$i_{\max}$	120	Iteration budget, beyond which the instance counts as unsolved
$n_{\mathcal{P}}$	175	Nonconvex QP instances in the test set
ϵ	10^{-4}	Floor on the corrected eigenvalues and pivots, shared by all strategies
β	0.5	Backtracking factor, $\alpha_{\text{prim}} \leftarrow \beta\alpha_{\text{prim}}$
$n_{\max}$	10	Maximum backtracking iterations
κ_s	10^{-1}	Switching constant, $\alpha_{\text{prim}}(-\rho) > \kappa_s\,\theta^2$
γ	10^{-4}	Armijo constant for the f-phase, $f^+ \leq f + \gamma\,\alpha_{\text{prim}}\rho$

5.4 Numerical comparison

In this section, we report the numerical results of the ten regularization methods on the nonconvex QPs of the benchmark. The comparison proceeds in three stages, each narrowing the field. We first compare the six Riccati-based methods against one another, in order to identify the strongest representatives of the family. These are then set against the full Hessian regularizations, and finally against inertia correction, which serves as the practical reference point.

5.4.1 Comparison among Riccati-based regularizations

We first compare the proposed Riccati-based regularization methods, which are the main focus of this work. As a baseline we also include the unregularized recursion (NO_REGULARIZE).

The iteration profile in Figure 5.2 reports, for each method, the number of instances solved within a given iteration budget, where two trends stand out. First, each PM dominates its counterpart M once the budget exceeds about 30 iterations, second, among the regularization strategies, MIRROR and GMW are clearly more effective than PROJECT. These trends validate the insights mentioned in Section

3.3. Among all the methods, PM_MIRROR tops the profile, solving 171 of the 175 instances, with PM_GMW close behind at 167.

Besides, it is interesting to note that the unregularized recursion can also solve a non-negligible fraction of the test set (31 of the 175 instances), which can be explained by the curvature contribution of the constraint that help to make the reduced Hessian positive definite in some instances.

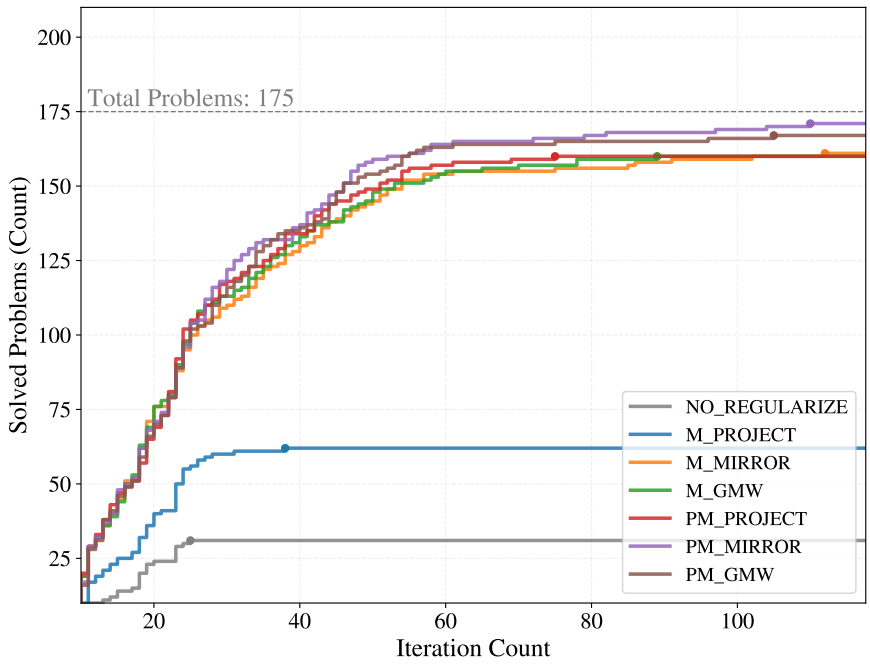


Figure 5.2: Iteration profile of the six Riccati-based methods over the 175-instance test set: number of instances solved within a given iteration budget.

Figure 5.3 complements this picture by showing the distribution of the worst-case decomposition cost in flop estimation, annotated with the maximum observed cost and the number of failures for each method. Since only the successful instances and maximal iteration instances are counted, the M_PROJECT will be excluded from the anaylsis, due to the frequent failing on 113 of the 175 instances.

In general, the PM incur a higher and more dispersed cost, reflecting the extra factorization work introduced by the cost-to-go term, yet they are generally more robust and thus the focus of the comparison. In Figure 5.3, PM_MIRROR is the most robust method with only 4 unsolved problems, and PM_GMW follows with 8 while keeping its worst-case cost well below that of both PM_MIRROR and PM_PROJECT. Notably, though the PM_PROJECT and PM_MIRROR belong to the same class of regularization strategy, the worst-case cost of PM_PROJECT is

higher than that of PM_MIRROR, which can be explained by the issue of inadequate regularization in the minimum perturbation strategy, so more frequency is needed in the PM_PROJECT case.

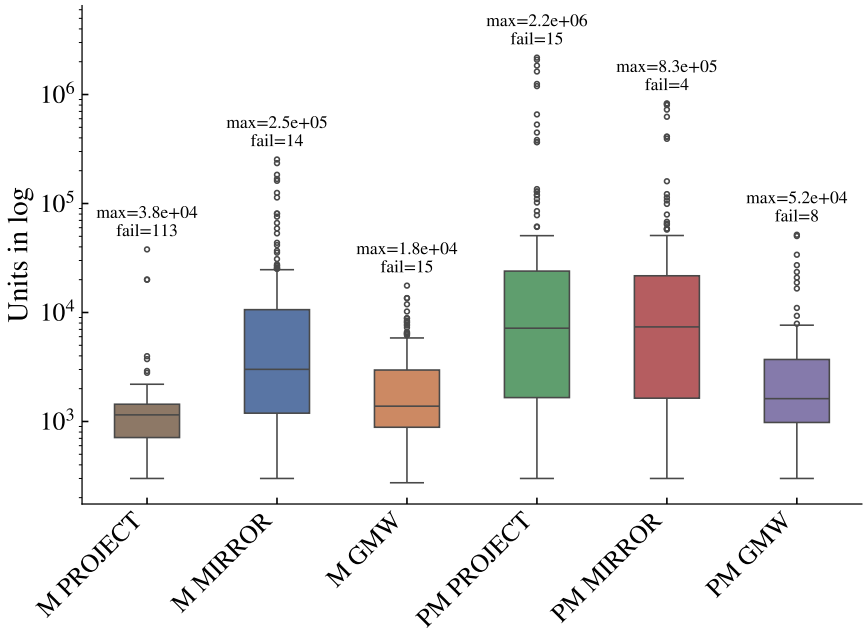


Figure 5.3: Worst-case decomposition cost of the six Riccati-based methods (flops, log scale). Annotations report the maximum observed cost and the number of unsolved instances.

Taken together, two figures first reveal a robustness ordering, PM_MIRROR > PM_GMW > {M_MIRROR, M_GMW, PM_PROJECT} > M_PROJECT, where PM_PROJECT only matches the plain M methods. If reliability is the priority, PM_MIRROR is the safest choice, as it leaves the fewest instances unsolved. If a balance between reliability and computational cost is preferred, PM_GMW is the more attractive option, which maintains the robustness (8 unsolved instances) at a much lower worst-case cost. We will therefore focus on PM_MIRROR and PM_GMW in the following comparisons with other regularization families.

5.4.2 Comparison with full Hessian regularizations

Having identified PM_MIRROR and PM_GMW as the strongest Riccati-based methods, we now compare them against the full Hessian regularizations.

The iteration profile in Figure 5.4 shows the two Riccati-based methods sitting well above every full-Hessian variant across the whole iteration budget.

PM_MIRROR and PM_GMW leave only 4 and 8 instances unsolved, whereas each H method leaves a substantial number of instances unsolved even at the largest budget. This suggests that regularizing the full Hessian does not recover the reliability or efficiency obtained by regularizing within the Riccati recursion.

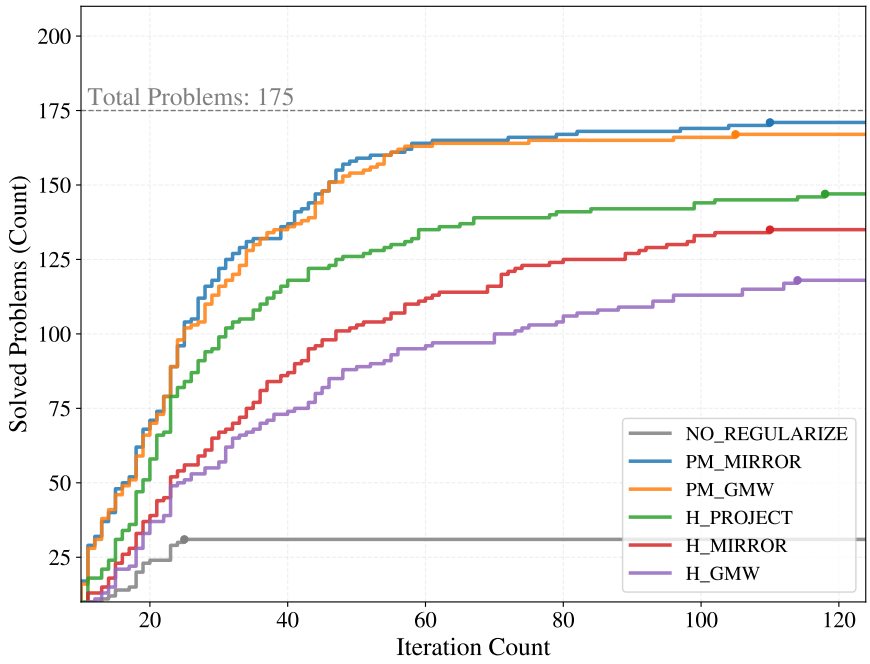


Figure 5.4: Iteration profile of the two strongest Riccati-based methods (PM_MIRROR, PM_GMW) against the full Hessian regularizations (H_PROJECT, H_MIRROR, H_GMW).

In Figure 5.5, the gap between Riccati-based methods and Full Hesian methods is even more pronounced, and it runs in the Riccati-based methods' favour on *both* axes. On the robustness, PM_MIRROR fails on only 4 instances and PM_GMW on 8, while the full-Hessian methods fail on 28 (H_PROJECT), 40 (H_MIRROR), and 57 (H_GMW) instances. On the cost, the full-Hessian variants are also more expensive: the worst-case costs of H_PROJECT and H_MIRROR lie far above those of both Riccati-based methods, and even the cheapest full-Hessian option, H_GMW, has a median cost well above that of PM_GMW, yet it is also the least robust of the methods compared here. The full Hessian methods thus are considered neither speed nor reliability relative to the Riccati-based approach.

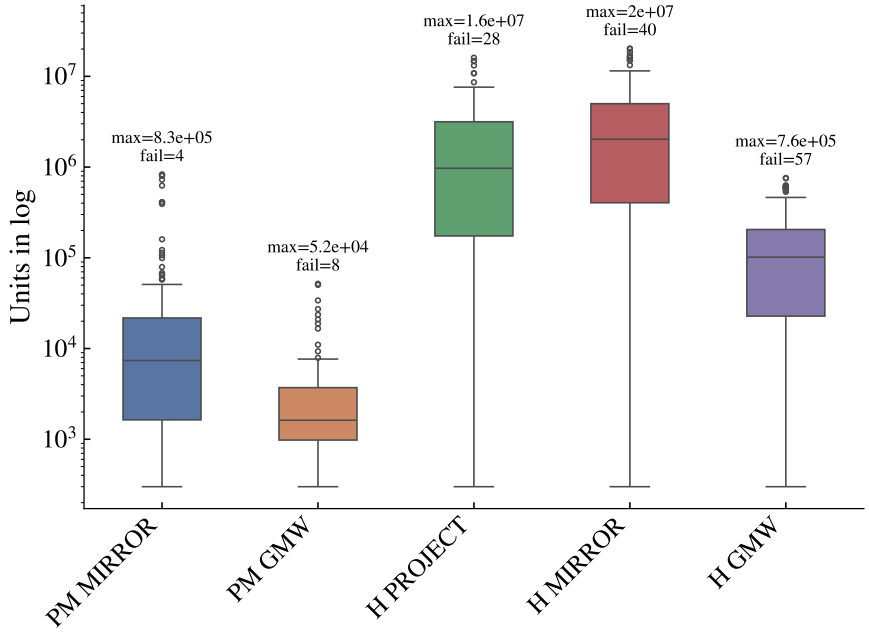


Figure 5.5: Worst-case decomposition cost of the two strongest Riccati-based methods against the full Hessian regularizations (flops, log scale). Annotations as in Figure 5.3.

The Riccati-based methods are simultaneously more robust and cheaper than every full-Hessian counterpart in Figure 5.5. This supports the central premise of the approach, namely that regularizing the reduced Hessian by exploiting the stagewise structure of the problem is preferable to regularizing the full Hessian.

5.4.3 Comparison with inertia correction

Finally, we compare the recommended Riccati-based methods against inertia correction (INERTIA_LM), which is widely used in practice and serves as a reference point for the comparison.

The iteration profiles in Figure 5.6 end close for the three regularized methods, with PM_MIRROR and PM_GMW solving the most instances (171 and 167) and INERTIA_LM close behind (164). However, beyond about 25 iterations the PM_MIRROR and PM_GMW sit clearly above INERTIA_LM, indicating less iteration is needed to solve the same number of instances.

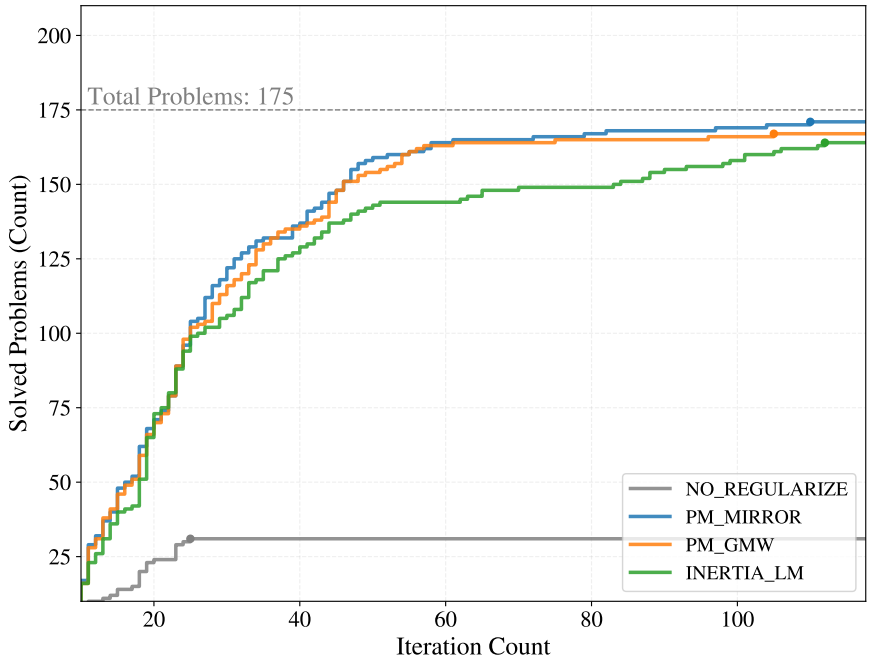


Figure 5.6: Iteration profile of PM_MIRROR and PM_GMW against inertia correction (INERTIA_LM).

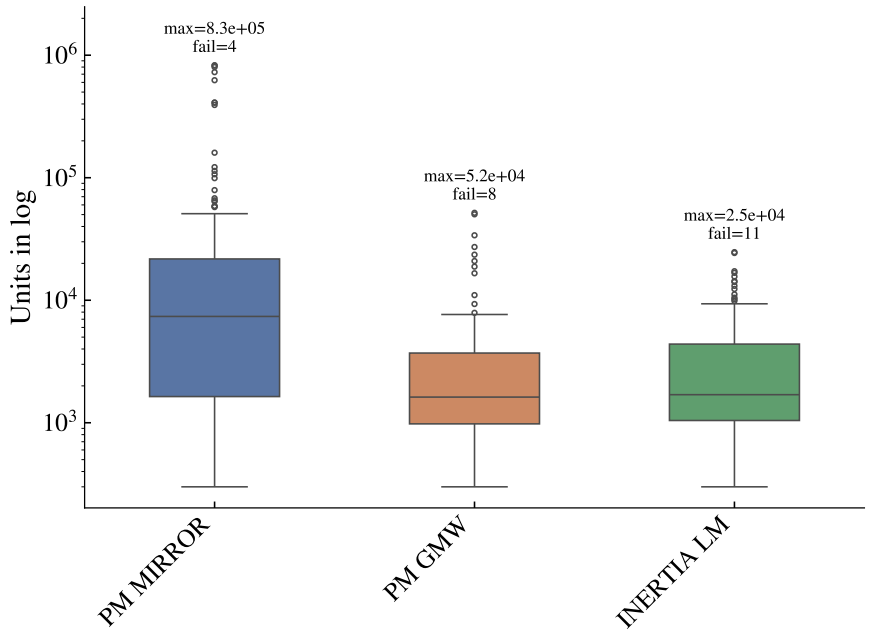


Figure 5.7: Worst-case decomposition cost of PM_MIRROR and PM_GMW against inertia correction (flops, log scale). Annotations as in Figure 5.3.

The decomposition-cost distribution in Figure 5.7 draws the distinction more

finely. Inertia correction is the cheapest of the three in the worst case, with a maximum cost of 2.5×10^4 flops, below both PM_GMW (5.2×10^4) and PM_MIRROR (8.3×10^5). In terms of median cost, however, INERTIA_LM and PM_GMW are essentially on par, with PM_GMW marginally lower. Their robustness differs more clearly, however: INERTIA_LM fails on 11 instances, against 8 for PM_GMW and 4 for PM_MIRROR.

From this benchmark, inertia correction is therefore preferable only where a hard bound on the per-decomposition cost outweighs the number of solved instances.

5.4.4 Ablation on the conditional Mehrotra

All results so far use Mehrotra's probing. To justify this choice, we rerun PM_MIRROR, PM_GMW, INERTIA_LM and H_PROJECT, the most robust full Hessian variant, together with the unregularized baseline, using the conditional predictor–corrector instead (`pred_corr`=`cond_pred_corr`= 1), with all other settings unchanged.

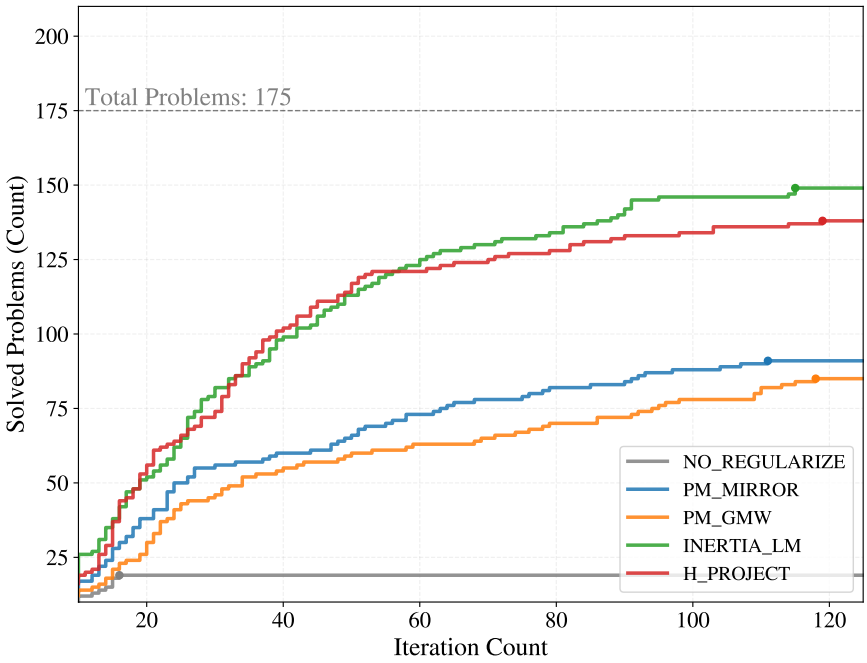


Figure 5.8: Iteration profile with the conditional predictor–corrector of PM_MIRROR, PM_GMW, INERTIA_LM, H_PROJECT and the unregularized baseline.

Figure 5.8 shows that every method solves fewer instances with the corrector,

but the loss is far from uniform. PM_MIRROR and PM_GMW drop from 171 and 167 to 91 and 85 solved instances, whereas INERTIA_LM and H_PROJECT only drop from 164 to 149 and from 147 to 138. The ranking of Section 5.4 is thereby reversed, and the two Riccati-based methods now trail both alternatives. All additional failures exhaust the iteration budget, and the worst-case decomposition cost in Figure 5.9 rises for every method, most visibly for PM_MIRROR, whose maximum grows from 8.3×10^5 to 2.2×10^6 flops.

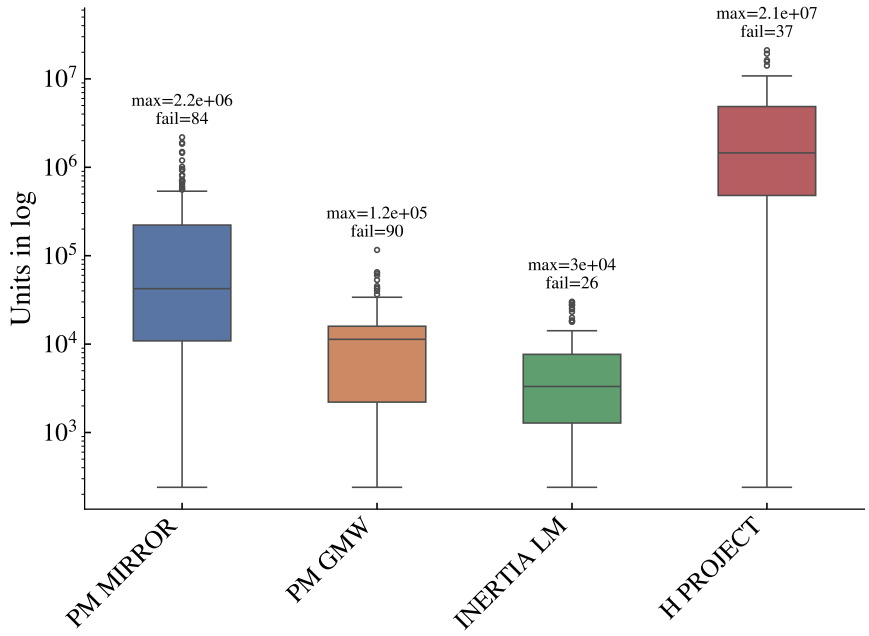


Figure 5.9: Worst-case decomposition cost with the conditional predictor–corrector (flops, log scale). Annotations as in Figure 5.3.

This is consistent with the argument in Section 2.3 and is why probing is adopted as the default barrier update in all other experiments. Section 5.5.3 discusses why PM_MIRROR loses far more instances than INERTIA_LM.

5.4.5 Ablation on the line search

The two-phase line search of Section 2.4.2 was introduced to safeguard the FTB step on nonconvex QPs. To isolate its contribution, we disable it, so that the FTB step is taken directly, and rerun the same five methods as in the previous subsection with all other settings unchanged.

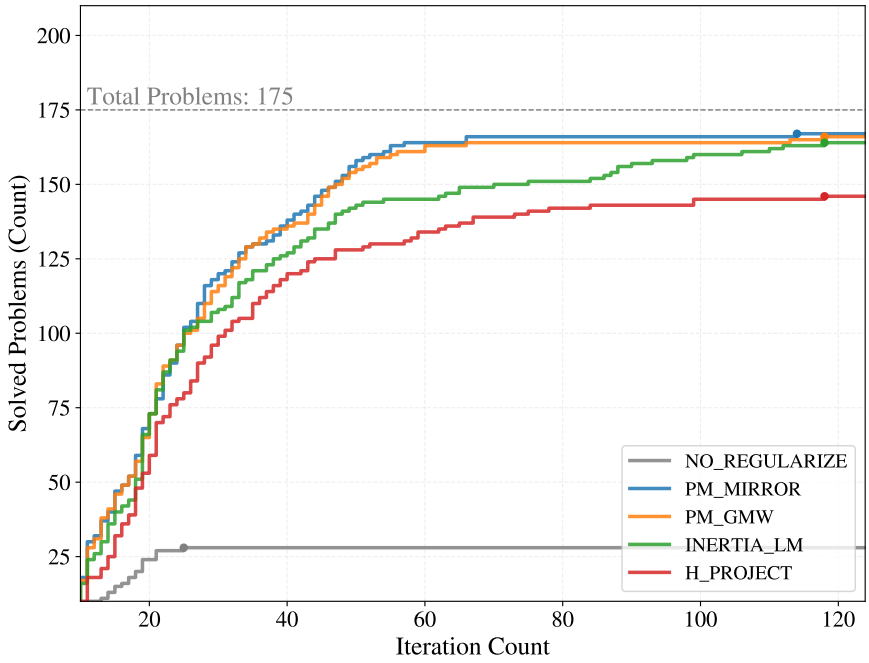


Figure 5.10: Iteration profile without the line search of PM_MIRROR, PM_GMW, INERTIA_LM, H_PROJECT and the unregularized baseline.

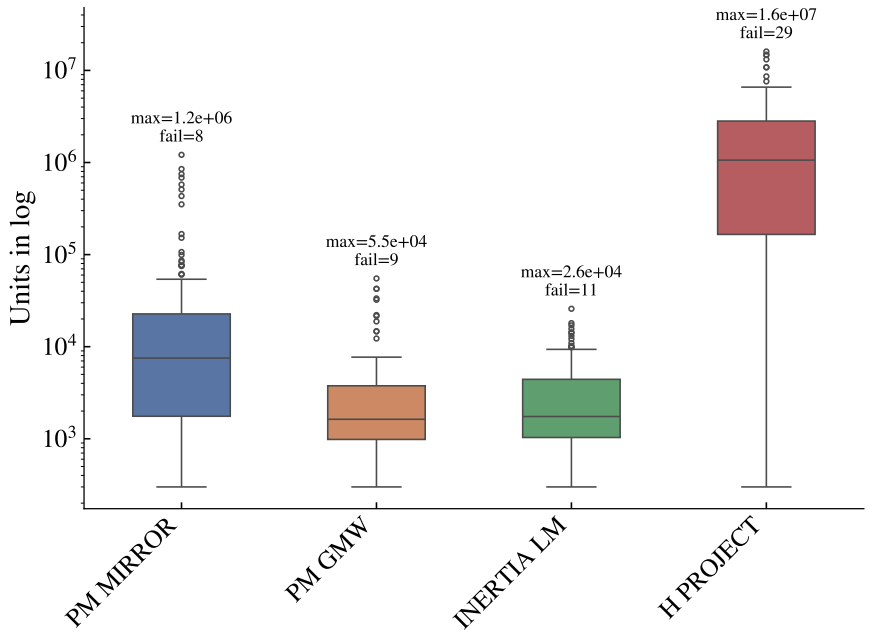


Figure 5.11: Worst-case decomposition cost without the line search (flops, log scale). Annotations as in Figure 5.3.

Figure 5.10 shows that the line search has a much smaller effect than the choice of barrier update. Without it, PM_MIRROR solves 167 instead of 171

instances, PM_GMW 166 instead of 167 and H_PROJECT 146 instead of 147, while INERTIA_LM solves 164 in both cases. PM_MIRROR benefits most, as the line search adds four solved instances and loses none, whereas for the other methods the instances it gains and loses nearly cancel. On the instances solved in both settings, most iteration counts are identical. The worst-case decomposition cost in Figure 5.11 also stays close to that with the line search, apart from the maximum of PM_MIRROR, which rises from 8.3×10^5 to 1.2×10^6 flops.

The ranking of Section 5.4 therefore does not rely on the line search. The robustness of the Riccati-based methods stems from the regularization itself, and Section 5.5.3 discusses why the line search mainly helps PM_MIRROR.

5.5 Discussion

The comparison in Section 5.4 counted the unsolved instances of each method. This section first resolves these failures along two directions. Section 5.5.1 splits them by the status with which the solve terminated, which separates a numerical breakdown of the regularized recursion from a mere exhaustion of the iteration budget. Section 5.5.2 splits them by the OCP each instance was collected from, which identifies what distinguishes the hard instances from the easy ones and explains why some methods perform poorly. Finally, Section 5.5.3 explains why the ablations of the barrier update and the line search affect PM_MIRROR and INERTIA_LM so differently.

5.5.1 Failure by status

A run that does not converge ends with one of three exit statuses, shown per method in Figure 5.12:

- *Unregularized*: the Cholesky factorization fails *again* after the regularization has been applied, usually caused by numerical indefiniteness.
- *Max iterations reached*: the residuals did not fall below their tolerances within the budget $i_{\max} = 120$.
- *NaN encountered*: an infinite or undefined value appeared in the residuals, so that the iterate itself has overflowed.

In Figure 5.12, the M_PROJECT is the only Riccati-based method with numerical breakdowns with 113 of its 175 instances unsolved. 110 of its 113 failures stem from numerical issue, while only 3 exhaust the iteration budget. This

confirms the analysis in Section 3.3, where ill-conditioning propagates into numerical indefiniteness or overflow. Adopting either remedy identified there resolves these breakdowns: abandoning the minimum-perturbation rule (M_MIRROR, M_GMW) or adding the cost-to-go correction (PM_PROJECT) eliminates all NaN and unregularized failures, leaving only iteration-limit terminations (14, 15, and 15 failures, respectively). Combining both remedies, as in PM_MIRROR and PM_GMW, further reduces the remaining iteration-limit failures. INERTIA_LM likewise fails only by reaching the iteration limit.

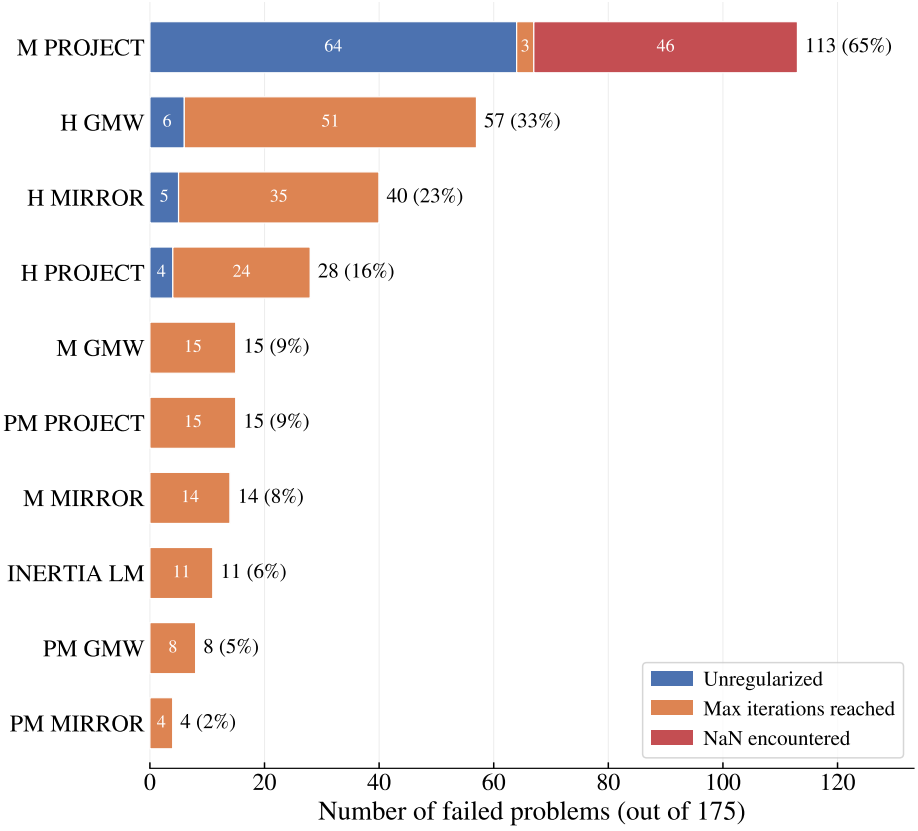


Figure 5.12: Failures of each method by exit status over the 175-instance test set. The bars are stacked by status and annotated with the total number of unsolved instances and the corresponding share of the test set.

The full Hessian methods break down on 4, 5 and 6 instances, but most of their failures reach the iteration limit (24, 35 and 51). Their breakdowns come from the cost-to-go matrix P_k, which no H strategy modifies. Near convergence, the barrier term $\Lambda_k T_k^{-1}$ grows as the slacks of active constraints approach zero, and the Riccati recursion carries this growth into P_k. Once $\kappa_2(P_k)\,\epsilon_{\text{mach}} \geq 1$, where ϵ_{mach} denotes the machine precision, rounding errors exceed the smallest

eigenvalue of P_k, and the computed P_k can become indefinite. The Riccati-based methods meet the same P_k but do not require $P_k \succ 0$, since they enforce $\tilde{M}_k \succ 0$ before each Cholesky factorization.

Accordingly, for the six methods that never suffer factorization breakdowns, namely the Riccati-based methods other than M_PROJECT together with INERTIA_LM, all failures reduce strictly to exhausted iteration limits, which plausibly reflect an inadequate regularization scale or an ineffective search direction.

5.5.2 Failure by category

Figure 5.13 resolves the same failures by the OCP each instance was collected from. The cell entries are the numbers of unsolved instances and the shading is the failure rate within the column, so that categories of different size remain comparable, and the eight columns are ordered by size and sum to the 175 instances of the test set.

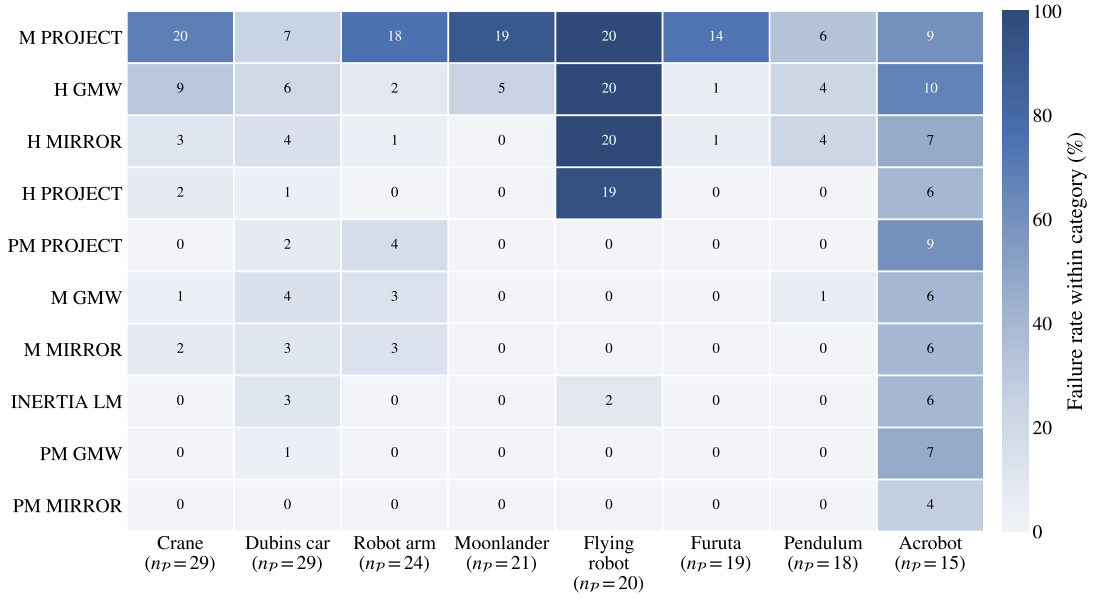


Figure 5.13: Failures of each method by OCP category. The cell entries give the number of unsolved instances and the shading gives the failure rate within the column, where $n_{\mathcal{P}}$ is the number of instances collected from that OCP. The rows are ordered by total failure count and the columns by size.

Comparing with Figure 4.1 suggests that an instance becomes hard when its indefiniteness is both deep and wide. Two *Crane* instances with very deep negative curvature are indefinite in only 5% and 7% of their stages, and every method

except M_PROJECT and H_GMW solves both: the isolated indefinite stage is corrected once and the recursion recovers. The four unsolved *Acrobot* instances instead have $|\lambda_{\min}| > 250$ in 33% to 53% of their stages, so large corrections are needed at many stages, and no method converges within the iteration budget. No other OCP in Figure 4.1 combines such depth with such a rate.

For the full Hessian methods, *Flying robot* is the hardest OCP: H_PROJECT, H_MIRROR and H_GMW fail on 19, 20 and 20 of its 20 instances, while every Riccati-based method except M_PROJECT solves all of them. In *Flying robot*, the indefiniteness lies only in the cost-to-go matrix and affects about 2% of the stages, while $R_k \succ 0$ throughout (Figure 4.4). The full Hessian methods cannot act on P_k, so they perturb $\tilde{Q}_k$ over the whole horizon, including the stages that need no correction, and exhaust the iteration budget. This supports the premise of Section 3.2 that indefiniteness should be treated where it arises in the recursion.

INERTIA_LM adds $\delta_w I$ to both $\tilde{Q}_k$ and $\tilde{R}_k$, which in the Riccati recursion shifts both P_k and $\tilde{M}_k$. It therefore solves 18 of the 20 *Flying robot* instances, and its total failure count is much closer to the PM methods than to the full Hessian ones. Unlike the Riccati-based methods, however, its shift is not localized: δ_w must be large enough for the most negative eigenvalue over all stages and is applied at every stage.

The figure also refines the M versus PM comparison of Section 5.4. M_MIRROR and M_GMW already solve every *Flying robot* instance. Apart from *Acrobot*, their failures lie mainly on *Dubins car*, *Crane* and *Robot arm*, two of which are dominated by the *both* case, where R_k and P_{k+1} are indefinite at the same stage. PM_MIRROR solves all instances of these three OCPs and PM_GMW fails on only one *Dubins car* instance, consistent with the propagation through P_k predicted in Section 3.3.

In summary, the instances that no method solves combine deep negative curvature with many indefinite stages, which in this benchmark occurs only in *Acrobot*. The full Hessian methods underperform because enforcing $\tilde{H}_k \succ 0$ alters the state curvature even when only a few stages are indefinite, as in *Flying robot*. Regularizing within the Riccati recursion avoids this by modifying only the indefinite matrices at the stages where they arise, although the *Acrobot* instances remain unsolved by all methods.

5.5.3 Analysis on ablation

Based on the two ablations, this subsection explains the behavior of PM_MIRROR and INERTIA_LM by addressing two questions:

1. Why does the ranking of Section 5.4 reverse under the conditional Mehrotra corrector?
2. Why does the line search mainly help PM_MIRROR but not INERTIA_LM?

Recall (2.9). With simplified notation, the regularized system reads

$$(\tilde{M}_i + E_i)\Delta y_z = -r_i, \tag{5.6}$$

and the step is bounded by

$$\|\Delta y_z\| \leq \frac{\|r_i\|}{\lambda_{\min}(\tilde{M}_i + E_i)}. \tag{5.7}$$

The bound holds for both methods, which differ in how small $\lambda_{\min}(\tilde{M}_i + E_i)$ can become.

INERTIA_LM increases a uniform shift δ until the factorization succeeds. If $\delta_{\min}$ is the smallest shift for which it succeeds, then $\lambda_{\min}(\tilde{M}_i + E_i) \geq \delta - \delta_{\min}$. Since the shift is increased by the factor κ_w^+, the accepted δ usually lies well above $\delta_{\min}$, and $\lambda_{\min}$ stays of the order of δ.

PM_MIRROR modifies only the negative eigenvalues of M_k and P_k, stage by stage. A small positive pivot passes the factorization unmodified and can produce large feedback gains, so nothing prevents $\lambda_{\min}(\tilde{M}_i + E_i)$ from approaching zero. On regularized iterations, PM_MIRROR reaches a much smaller $\lambda_{\min}(\tilde{M}_i + E_i)$ than INERTIA_LM at similar $\|r_i\|$, and its affine steps are correspondingly longer, thus a full affine step could be impossible to be performed.

On the Barrier Update Strategies. Mehrotra's corrector adds the second-order term $\Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1}$ to the right-hand side, which grows quadratically with the affine step. As shown in Table 5.4, the median affine step of PM_MIRROR is far longer than that of INERTIA_LM.

In PM_MIRROR, the second-order term then dominates the right-hand side and results a long corrector step, the fraction-to-boundary rule, thus, leaves only a very short step size. Since, the affine step is itself cut short, μ_{aff} stays close to μ, and the very short corrected step leaves μ_{cor} close to μ as well, below $2\mu_{\text{aff}}$. The safeguard of Section 2.3 rarely detects this, so the bad corrector step is taken,

which is dominated by a correction for the second-order error of an affine step that cannot be taken, so it neither reduces the residuals nor moves the iterate along the directions needed to leave the indefinite region.

INERTIA_LM lifts every direction by the uniform shift δI, which keeps $\lambda_{\min}$ away from zero and the affine step moderate. The second-order term therefore stays comparable in size to the rest of the right-hand side, so the corrected direction remains moderate as well, and the fraction-to-boundary rule leaves a step size comparable to that of a centering step. Since μ_{aff} then lies clearly below μ, the safeguard is effective and rejects 39.1% of the corrector steps. The conditional corrector step thus makes progress as intended, and INERTIA_LM loses far fewer instances to the corrector than PM_MIRROR.

With probing, the affine step only determines the centering parameter, so the second-order term no longer enters the search direction, and the ranking reverses. The shift δI in INERTIA_LM raises the curvature of directions that are already positive, which shortens the step along weakly curved directions. INERTIA_LM therefore moves less per regularized iteration and needs more of them, as one can see in Table 5.4. PM_MIRROR, in contrast, keeps its steps along weakly curved directions long. Without the second-order term, these long steps no longer distort the search direction, so PM_MIRROR moves further per regularized iteration, leaves the region where the reduced Hessian is indefinite sooner, and reaches the regime where unregularized Newton steps converge quickly.

Table 5.4: Step metrics of PM_MIRROR and INERTIA_LM.

	PM_MIRROR	INERTIA_LM
Corrector, on 175 *instances*		
Median affine step $\|\Delta y_z\|$	65.4	4.4
Percentage of rejected corrector steps	3.7 %	39.1 %
Probing, on 164 *instances solved by both*		
Iterations	4218	5132
Regularized iterations	1566	2289
Median displacement $\alpha_{\text{prim}}\|\Delta y_z\|$	1.44	0.47

On the Line Search. The two-phase line search distinguishes three situations:

1. high infeasibility or an ascent direction, handled by an h-phase step;
2. moderate or low infeasibility, handled by an f-phase step;
3. feasibility, handled by an f-phase step.

Since the constraints are affine, every step scales the violation as $\theta^+ = (1-\alpha_{\text{prim}})\,\theta$, so the line search never has to enforce feasibility. Backtracking in the f-phase is triggered only by the barrier objective, when the fraction-to-boundary rule drives a slack close to its bound and the barrier increase exceeds the predicted decrease $\alpha_{\text{prim}}|\rho|$ of a short step.

In PM_MIRROR, the short step size corresponding to the long regularized directions barely reduce θ, so PM_MIRROR remains in situation 2 throughout its regularized iterations. A trial step there predicts little decrease, so the Armijo condition rejects it, and backtracking keeps the blocking slack off its bound.

INERTIA_LM shifts the curvature of all directions uniformly, which keeps its directions short, so the fraction-to-boundary rule rarely cuts them. The predicted decrease of these long steps outweighs the barrier increase, so the f-phase rarely backtracks, and the instances the line search gains and loses for INERTIA_LM cancel.

Chapter 6

Conclusion and outlook

This thesis regularized the indefinite QPs of exact-Hessian SQP for NMPC inside the interior point solver, where the barrier curvature is already included and only the reduced Hessian has to be positive definite. Since the Riccati backward pass is a block $LDL^\top$ factorization of the reduced Hessian, indefiniteness is detected and corrected stage-wise wherever the Cholesky factorization of $\tilde{M}_k$ fails, and the recursion stays linear in the horizon length.

We showed that a minimum-perturbation correction of $\tilde{M}_k$ adds a negative rank-one term to the cost-to-go matrix, which can make the preceding stages indefinite and ill-conditioned. The improved strategy therefore uses a non-minimal correction and also regularizes the cost-to-go matrix. We proved that the regularization becomes inactive close to the solution, so the local convergence rate of the interior point method is retained.

In the benchmark of 175 indefinite QPs from eight OCPs, the cost-to-go matrix is indefinite in most indefinite stages. With Mehrotra's probing, PM_MIRROR left 4 instances unsolved and PM_GMW 8 at a lower worst-case decomposition cost than PM_MIRROR, against 28 to 57 for full Hessian regularization and 11 for inertia correction. The full Hessian methods failed on almost every *Flying robot* instance, where only the cost-to-go matrix is indefinite, and only at a few stages. The instances that no method solved combine deep negative curvature with a large share of indefinite stages.

Several questions remain open:

- **Regularization magnitude.** The eigenvalue floor ϵ is fixed, and the remaining failures occur where large corrections are applied at many stages. Adapting the magnitude, e.g. to the conditioning of the corrected blocks or to the barrier parameter, could improve robustness.
- **Other MPC formulations.** In time-optimal and economic MPC, the cost

provides little positive curvature, so indefiniteness is likely to be broader than in this benchmark. Whether the ranking of the methods carries over remains to be tested.

- **Embedding in SQP.** The QPs were solved in isolation with a Python prototype. An implementation in HPIPM and `acados` is needed to compare against SQP-level regularization in terms of SQP iterations, run time and closed-loop performance.

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